

Make, Buy, or Share: An Empirical Model of Wastewater Management in the Shale Gas Industry

Matthew O’Keefe*

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Abstract

Fracking has caused an increase in oil and gas wastewater production throughout the United States. Conventional injection well disposal poses risks to the environment. A promising alternative is reuse. I build and estimate an empirical model of wastewater management in which firms substitute between injection disposal and reuse. I use the model to quantify the effects of existing and proposed injection disposal taxes. The analysis reveals evidence of substantial frictions that raise the cost of reuse by inhibiting the reallocation (“sharing”) of reusable wastewater across firms. I investigate the nature of these frictions and discuss implications for wastewater policy.

Keywords: fracking; produced water; injection well disposal; boundary of the firm; transaction costs; matching

*Vanderbilt University. Email: matthew.okeefe@vanderbilt.edu. I am indebted to Rob Porter, Mar Reguant, and Vivek Bhattacharya for generous support and guidance. Michael Powell, Gaston Illanes, Bill Rogerson, Tom Hubbard, Igal Hendel, Meghan Busse, Ryan Kellogg, Alex MacKay, Ameet Morjaria, Lynne Kiesling, Rick Bond, and many more individuals provided insightful comments on earlier drafts of this paper. I acknowledge helpful comments from seminar participants at Northwestern, the Dallas Fed, UC Berkeley ARE, Johns Hopkins, DePaul, Vanderbilt, and UT Austin. Any errors are my own.

1 Introduction

Oil and gas wells produce significant quantities of hazardous wastewater in addition to hydrocarbons. The widespread adoption of hydraulic fracturing (“fracking”) in recent decades has caused a sharp increase in wastewater volumes across the US, as wells which have been fracked tend to produce more wastewater than conventional oil and gas wells.

Wastewater disposal is often achieved through the use of specialized “injection disposal” wells that sequester fluids deep underground. In recent years, growing concern over potential environmental and economic impacts from injection disposal have lead regulators in key oil and gas producing states including Oklahoma (Wertz, 2014; Money, 2019), New Mexico (Bennett, 2023), and Texas (Gough, 2025), to restrict permitting and/or daily injection rates for injection wells. Ohio and New Mexico have levied volumetric taxes on injection disposal. While often considered “safer” than alternatives such as surface water discharge or evaporation, injection disposal can cause seismic activity and creates risks of groundwater contamination and surface spills (McCurdy, 2011; Groundwater Protection Council, 2019). Economically, subsurface pressure from injection disposal has the potential to raise local drilling costs (Bruant, 2023) and reduce productivity for nearby wells (McFarland, 2025).

One important alternative to injection disposal is wastewater *reuse*. For many fracking operations, minimally treated wastewater can directly substitute for freshwater that would otherwise be drawn from nearby ground or surface water resources. Reuse is privately efficient whenever the costs of transporting and treating water for reuse fall below the combined costs of freshwater acquisition and conventional disposal (Veil, 2012). From an environmental perspective, reuse usefully mitigates water-related externalities (Cooper et al., 2022). The Pennsylvania Department of Environmental Protection (DEP) “promotes [reuse] as a first option when it is technically and economically feasible” (PA DEP, 2024).

Despite continued growth in wastewater volumes, little work in economics to date has explored the significance of wastewater reuse for the optimal regulation of injection disposal, or indeed the optimal regulation of oil and gas wastewater more generally.¹ The first contribution of this paper is an empirical model of oil and gas firms’ water and wastewater management decisions in which firms can substitute between injection disposal and reuse. The basic framework that I develop can be used to estimate the costs of reuse and to simulate the equilibrium impacts of alternative wastewater regulations, as illustrated below.

A second contribution is to establish the empirical significance of transaction costs that inhibit the transfer of reusable wastewater between firms. The importance of transaction costs

¹Wastewater management has been the subject of a sizable engineering literature, including Puder and Veil (2006), Acharya et al. (2011), Karapataki (2012), Veil (2012), and Hussey and Burnett (2017).

vis à vis injection disposal follows from an inherent asymmetry between water consumption in fracking and wastewater production. New completions consume vast amounts of water, while mature wells continuously produce smaller and ever-declining quantities of wastewater roughly in proportion to hydrocarbons. Efficient reuse therefore entails a many-to-one matching process in which wastewater from the large stock of mature wells is transported to a small number of ongoing completions. Over time, individual firms complete new wells at varying rates and in different locations in response to idiosyncratic shocks. As a result, there are frequently transportation cost savings from re-allocating wastewater across firms. Yet if re-allocation across firms (or “sharing”) is costly, firms may choose injection disposal even when reuse would generate cost savings under horizontal integration.

In the model, firms choose the least cost destination for each barrel of wastewater and, simultaneously, the least cost source for each incremental barrel of base fluid for use in fracking (either freshwater or wastewater). In the case of sharing transactions, firms incur transaction costs and make transfer payments that reflect the equilibrium division of surplus between senders and receivers of wastewater. Formally, I adopt a transferable utility (TU) matching framework (Choo and Siow, 2006; Galichon and Salanié, 2022). In equilibrium, all firms minimize total costs taking transfer payments as given, and producer surplus is maximized conditional on transaction costs. Under specific parametric assumptions on the distribution of unobserved costs the model yields a gravity equation.² Parameters governing producer surplus and transaction costs can then be estimated by Poisson regression.

I estimate the model using wastewater disposition records for unconventional oil and gas wells in Pennsylvania from 2017-2020. Crucially, the records include both within and between-firm shipments of wastewater, facilitating the identification of transaction costs. Following rapid increases in unconventional drilling, Pennsylvania became the second largest producer of natural gas in the United States in 2013. This growth occurred in spite of a dearth of in-state injection well capacity. In the data, 90% of wastewater is reused. Sharing appears to play a meaningful role in sustaining this level of reuse: 8% of wastewater is transferred to facilities operated by rival oil and gas producers. As nearly all significant firms are party to some apparent sharing transactions (either as sender or receiver), I am able to allow for considerable shipment-level heterogeneity in transaction costs during estimation.

I find that reuse in general generates significant producer surplus, but much of this surplus is lost in the case of sharing transactions. In the most parsimonious specification I consider, transaction costs are equivalent in magnitude to the marginal cost of shipping a tanker truck of wastewater more than one hundred additional miles. This estimate blends extensive and intensive margin frictions: many firms that could profitably share do not, while firms that

²The connection between TU matching and gravity models is recognized in Galichon and Salanié (2024).

share often fail to do so efficiently. Further results illustrate that transaction costs can vary considerably across a firm’s relationships and are poorly predicted by observables such as firm size and geology. These findings are suggestive that some degree of misallocation occurs in practice, including the possibility of excessive injection disposal.³

A small number of injection wells for oil and gas wastewater operate in the state of Pennsylvania. However, the large majority of Pennsylvania wastewater that is not reused is transferred to injection wells in neighboring Ohio.⁴ Ohio has charged a \$0.20 per barrel tax on wastewater disposal since 2010. To demonstrate the usefulness of the empirical framework for policy analysis, I quantify the impact of this tax. My preferred estimates suggest that the tax reduced Pennsylvania injection disposal by only 1.4%. After unexpected seismic activity in 2011 near Youngstown, OH was attributed to nearby injection disposal, Ohio Governor John Kasich proposed to raise the state’s disposal tax to \$1.00 per barrel. I predict that this alternative policy would have reduced Pennsylvania injection disposal by only 6.2%. While sensitive to modeling assumptions, these results suggest that the demand for injection disposal is inelastic, and that appreciably larger disposal taxes may be required if Ohio policymakers wish to significantly reduce injection volumes below current levels.

Equilibrium models in empirical microeconomics often begin from an assumption of efficient, frictionless trade. One important conclusion from the policy analysis is that, in the context of wastewater management, failing to account for transaction costs can lead to incorrect estimates of policy impacts. In particular, I find that a similar model which did not account for transaction costs would significantly overstate the impacts of both the existing and proposed Ohio disposal taxes. This occurs because potential sharing transactions tend to require longer shipments than internal reuse. If one assumes that observed shipments arise in a frictionless equilibrium, omission of transaction costs causes an upward bias in the relative cost of distance, and hence the elasticity of demand for injection disposal.

As in other contexts, a “portfolio” approach to policy can be useful when externalities interact with other market imperfections (Jaffe et al., 2005). Standard Pigouvian analysis suggests that taxes on injection disposal and restrictions on the permitting and operation of injection wells (i.e., quantity restrictions) are appropriate policy responses to an underpricing

³“Misallocation” refers here to the distortion relative to the outcome attained under complete integration. This type of misallocation is not necessarily inefficient if transaction costs are *welfare-relevant*. The distinction between welfare-relevant and welfare-irrelevant costs appears in the literature on switching costs in consumer markets (e.g., Handel and Kolstad, 2015). See also Buchanan and Stubblebine (1962).

⁴The scarcity of injection disposal capacity in Pennsylvania has been attributed to unfavorable geology and a relatively difficult permitting process (McCurdy, 2011). To prevent fluid migration, injection wells target high porosity rock formations (e.g., sandstone) sealed beneath non-porous formations (e.g., shale). To date few suitable locations have been identified. For historical reasons, Pennsylvania injection wells require EPA permits that are not needed for wells in most oil and gas producing states (Quigley, 2016).

of injection disposal.⁵ I conclude the paper by showing via counterfactual simulation how policies aimed at reducing transaction costs in wastewater sharing could indirectly reduce injection disposal. Two areas of opportunity that I highlight are greater public disclosure of information about wastewater disposition and greater legal and regulatory certainty around wastewater ownership and liability (Groundwater Protection Council, 2019).

From another perspective, the central observation of this paper is that the extent of reuse can be sensitive to the organization of production: the allocation of control rights for productive assets (i.e., wells) and the nature of vertical relationships among firms. The empirical framework presented here might be viewed an effective but fundamentally backwards-looking strategy for taking these elements into account when conducting policy analysis. In reality, the organization of production can change quickly: corporate transactions are frequent in the US oil and gas industry, while vertical relationships are not necessarily stable over time. One implication of the analysis is that policy-relevant elasticities are sensitive to such changes, implying that even the short-term effects of corrective taxation might be difficult to forecast using traditional methods. By the logic of Weitzman (1974), these considerations perhaps strengthen the case for quantity-based regulation of injection disposal.

The remainder of the paper is structured as follows. This section continues with a discussion of prior literature. Section 2 introduces the data and provides descriptive evidence on reuse, sharing, and transaction costs. Sections 3 and 4 present the model and estimation method, respectively. Section 5 analyzes the estimates, focusing on transaction costs and producer surplus. Section 6 analyzes the Ohio disposal tax. Section 7 discusses potential policy responses to the presence of transaction costs. Section 8 concludes.

1.1 Related literature

A large applied literature studies local impacts of the Shale Boom, including water-related impacts such as groundwater contamination (Muehlenbachs et al., 2015; Hill and Ma, 2022) and externalities related to the transport of water by truck (Muehlenbachs et al., 2021).⁶ I add to this literature by showing how local impacts can be mediated by the organization of production. This recognition highlights potential challenges for policy evaluation, but may also expand the set of policy instruments available to local policymakers.

I contribute to a small literature in industrial organization that applies TU matching models to analyze transactions between firms (Fox and Bajari, 2013; Akkus et al., 2016; Fox, 2018). The models in these papers relax Choo and Siow (2006)’s assumption that unobserved heterogeneity is additively separable across agents but require maximum score estimation.

⁵I do not attempt to estimate the external costs of injection disposal in this paper.

⁶Jackson et al. (2014), Mason et al. (2015), and Black et al. (2021) offer comprehensive surveys.

TU matching models that retain additive separability have been estimated in the context of marriage (e.g., Chiappori et al., 2017) and labor markets (e.g., Corblet, 2025).

The estimation strategy closely resembles that of Atalay et al. (2019), who also rely on spatial variation to quantify firm boundaries using a gravity model (albeit one derived from very different microfoundations). As in that paper, transaction costs are identified by the relative probability of observing a shipment within versus between firms at a given distance. I reach a similar conclusion that the costs of crossing the firm boundary (here, the transaction costs of sharing) are frequently large in comparison to gains from trade. Atalay et al. (2019) analyze a large swathe of goods-producing and goods-distributing firms in the United States using Census data, focusing on heterogeneity across sectors. The wastewater setting affords the opportunity to more closely examine the extent to which the same features can vary across relationships within a relatively homogeneous set of transactions.⁷

Several authors in industrial organization have examined oil and gas productivity after the Shale Boom. Covert (2015) and Steck (2022) estimate learning models with the aim of understanding shale producers’ input choices. Agerton (2020) studies location choice. These papers each recognize water as a newly important input after fracking. Yet none directly addresses how water is sourced or how wastewater is optimally managed. I show that these decisions have important implications both for producer surplus and for the environment.

2 Background

This section introduces the data and briefly describes current oil and gas wastewater management practices in Pennsylvania. I also present descriptive evidence of transaction costs.

The primary data are derived from monthly disposal records that oil and gas firms file with the Pennsylvania DEP. These records indicate the disposal method and destination of all quantities of waste materials leaving each well pad, including every barrel of wastewater.⁸ Only quantities are recorded: I observe no information relating to prices or other contract terms. However, transfers for reuse are clearly indicated. Beginning in 2017, information identifying the destination facility is included in the case of transfers for reuse. I therefore focus on the period from 2017 to 2020. During this period, natural gas prices were generally

⁷Atalay et al. (2019) frame their results as the first estimates of the “actual magnitudes of the net benefits of internal transactions,” contrasting their structural approach with a large body of older empirical work testing comparative statics from the theory of the firm. From this perspective, evidence on the extent of within-market heterogeneity in transaction costs is novel to the literature on the theory of the firm.

⁸A well pad is a small parcel of land from which multiple wells (typically less than ten) are drilled in different directions or at different depths. Wells on the same pad share common infrastructure such as access roads and (importantly) wastewater storage tanks, making well-level waste attribution challenging.

low and investment in Appalachia began to weaken from its Shale Boom peak.

2.1 Injection disposal vs. reuse

Oil and gas extraction in Pennsylvania is conducted by numerous firms ranging from small, independent firms operating only a few wells to the largest global energy firms (Small et al., 2014). Figure 1 shows the locations of well pads operated by each of the twenty largest operators (by disposal volume) in the period that I study. The clustering visible in the figure reflects economies of density in exploration, leasing, drilling, and marketing, as well as in freshwater and wastewater management, which I discuss in this section.

During fracking, water is blended with sand or other small particles and specialized chemicals before being injected into a well under pressure. In the period that I study, a typical “completion” required more than one hundred thousand barrels of water.⁹ Once a well begins to produce hydrocarbons, a large share of this fluid returns to the surface along with significant volumes of groundwater. Wastewater production continues for the life of a well in steadily diminishing volumes.¹⁰ For a typical Marcellus well, lifetime wastewater production amounts to around 50% of fracking fluid volume (Kondash et al., 2018).¹¹

Wastewater is highly saline and may contain organic compounds, metals, and naturally occurring radioactive materials from groundwater in addition to fracking chemicals. Careful handling and specialized disposal are required by law (Groundwater Protection Council, 2019). Firms in Pennsylvania generally manage wastewater through a combination of reuse and injection disposal.¹² In either case, the vast majority of wastewater is transported by truck.¹³ Injection disposal involves the use of specialized wells to deposit wastewater deep underground beneath impermeable rock formations. Reuse entails transporting wastewater to a new well pad, performing inexpensive treatments such as filtering and pH reduction, and then using this treated water in place of freshwater when fracking new wells. In some cases treatment occurs at a dedicated site known as a centralized treatment facility (CTF)

⁹A “barrel” is a standard measure of liquid volume in the oil and gas sector. A barrel holds 42 gallons (159 liters). A single barrel is abbreviated “bbl” and 1,000 barrels “Mbbbl.”

¹⁰Water produced during or immediately after completion is often referred to as “flowback,” while water produced later is more commonly known as “produced water.” I use the term wastewater to encompass both.

¹¹In this respect the Marcellus region is quite different from other major shale plays in the United States such as the Permian in New Mexico and Texas. In “wet” shales, produced water volumes can easily exceed several multiples of fracking fluid volume (Kondash et al., 2018). Consequently, the maximum feasible rate of reuse outside Appalachia can be much lower than what is observed in Pennsylvania.

¹²In western states, evaporation and agricultural application are also used (Groundwater Protection Council, 2019). Another less commonly used option is desalination. In Pennsylvania, reuse outside the oil and gas industry is extremely limited. Pennsylvania effectively prohibited surface water discharge in 2011.

¹³Approximately 5% of produced water in Pennsylvania is transported by pipeline (Groundwater Protection Council, 2019). According to the DEP, some operators have also used rail transportation.

rather than taking place directly on a well pad, though the latter is more common.¹⁴

To contextualize later estimates, I note that disposal well operators typically charge fees on the order of \$1.00 per barrel of wastewater (Puder and Veil, 2006; Tavakkoli et al., 2020), while simple treatments for reuse cost \$0.50-\$1.00 per barrel (Oklahoma Water Resources Board, 2017). The all-in cost of freshwater (inclusive of pumping and transport) is generally less than \$0.25 per barrel (Acharya et al., 2011).

Due to favorable geology and ease of permitting, the large majority of injection wells in Appalachia are located in eastern Ohio. The distance between Pennsylvania gas wells and Ohio injection wells can be significant. This is illustrated in Figure 2, which shows the locations of injection wells relative to gas wells observed in the data. The dense cluster of drilling activity in the northeast is roughly 300 miles from Ohio injection wells. The cost of shipping wastewater such a distance via truck can easily exceed \$5.00 per barrel.¹⁵

Table 1 summarizes the average monthly disposition of wastewater reported in the data, aggregating across all firms and well pads in Pennsylvania over the analysis period. 88.7% of wastewater is transferred to a Pennsylvania well pad or CTF, while only 8.1% of wastewater is transferred to an injection well. A small amount of wastewater is transferred to other destinations, primarily well pads and CTFs in West Virginia and Ohio.

Table 2 disaggregates the above figures according to the location and size of the reporting firm. There are two major gas-producing regions within Pennsylvania, which can be seen in Figures 1 and 2: one in the southwestern part of the state, and one in the northeast. I first assign each firm to the either “Southwest,” “Northeast,” or “Peripheral” regions.¹⁶ Within the first two regions, I separate the five largest firms (“Large”) from all remaining firms (“Small”) based on natural gas production. Due to the significant transportation costs associated with injection disposal, firms in the Northeast report much less injection disposal (3.3% vs. 7.5% for firms in the Southwest).¹⁷ Injection disposal is also less common for Large firms than for Small firms (3.1% vs. 11.9%). The extent of heterogeneity within each region is illustrated in Figure 3d, which depicts injection disposal rates by well pad. Injection disposal is most common near the Ohio border.

¹⁴The choice between CTF and on-pad treatment depends on the tradeoff between scale economies in treatment and transportation costs. The use of CTFs can increase transportation costs because wastewater must be transported twice – once to the CTF, and then again to a location where it can be reused. There are also differences in regulatory treatment (such as bonding requirements).

¹⁵I discuss the cost of trucking in greater depth in Section 4.

¹⁶Firms are assigned to regions as follows. First, I assign each well pad to one of the six Assessment Units from the USGS National Assessment Project for oil and gas resources (Higley et al., 2019). Next, I map Assessment Units to the Southwest, Northeast, and Peripheral regions as shown in Figure 3a. Finally, I assign each firm to a region using the region in which the plurality of a firm’s well pads are located.

¹⁷For Peripheral firms, injection disposal is the predominant method of disposal (86.8%). The high rate of injection disposal among this group is largely driven by a single firm.

2.2 Wastewater sharing

Because wastewater production occurs gradually over the life of a well, the number of wells generating wastewater (a stock) is large in comparison to the number of new completions (a flow). Consequently, reuse entails a many-to-one matching process. To illustrate this point, Table 2 summarizes the number of facilities appearing in the data each month and associated shipment volumes (in “truckloads” of 110 barrels).¹⁸ In the average month, 1,712.6 distinct well pads reported wastewater disposal, while only 55.0 total completions occurred.¹⁹ The median monthly disposal volume was only 3.7 truckloads — less than 0.5% of the fluid volume needed to complete a new well. Conversely, facilities which appear as destinations tend to receive wastewater from many distinct sources: a well pad appearing as destination received 430.8 truckloads of wastewater from 31.4 distinct well pads on average.

Due to this asymmetry, firms can potentially benefit from trading wastewater when imbalances arise or when geographic synergies emerge. In Table 1, I classify well pad and CTF destinations as “internal” or “rival” based on the identity of the primary operating firm associated with each facility, where a “rival” refers to another oil and gas producer. CTFs operated by third party treatment firms are designated as “third party.”²⁰ Although the large majority of wastewater is transferred to internal or third party facilities, 9.4% is transferred to well pads and CTFs operated by rival firms. I interpret this fact as evidence that sharing plays an important role in explaining observed patterns of reuse.

For the remainder of the paper a “sharing transaction” is defined as a shipment to a rival well pad and CTFs. This definition is admittedly imprecise due to an important limitation of the data. When wastewater is initially transferred to a CTF, the ultimate location of reuse is not reported. Although the DEP prohibits “re-transfers” of wastewater sent directly to particular well pads, this is not the case for shipments to CTFs. Thus, wastewater might in principle be transferred from Firm A’s well pad to Firm B’s CTF for processing before being re-transferred to Firm A’s well pad for reuse. For the purpose of the remaining analysis I assume this never occurs. I further assume that wastewater initially transferred to internal or third party CTFs is never re-transferred to a rival firm.²¹

¹⁸The raw data are reported in barrels. The modal shipment volume in the data is 110 barrels. In practice, water-hauling tanker truck capacity varies from about 80 to 130 barrels.

¹⁹Not reported in Table 2. Calculation based on the average number of fracking jobs recorded in FracFocus during the sample period. By comparison, EIA’s Drilled but Uncompleted Wells (DUC) data implies 101.5 completions per month for the whole of Appalachia, including Ohio and West Virginia.

²⁰Producer-affiliated CTFs are often little more than semi-permanent systems of tanks or impoundments where the same treatments conducted on a well pad can be conducted at a larger scale. Third party CTFs are constructed similarly but may also have technologies that can treat water to higher standards, although these technologies are rarely used in practice.

²¹Without these assumptions, the “true” extent of sharing might vary from as little as 6% to as much as 42% of total disposal. The upper bound would require that 100% of transfers to a firm’s own CTFs are

2.2.1 Which firms share wastewater?

During the analysis period, 49 out of 75 operating firms shared wastewater (i.e., transferred wastewater to a rival well pad or CTF) on at least one occasion, including 9 of the 10 largest firms by wastewater production. On average, a firm that shared at least once shared in more than half of all months. Thus, most firms share and many do so often.

Table 2 disaggregates the overall sharing rate by firm location and size. Sharing is more frequent among firms in the Northeast (13.7% vs. 6.4% in the Southeast) and among Small firms (33.7% vs. 3.6% for Large firms). For firms in the Northeast, the main alternative to sharing (injection disposal) is much more costly on account of the greater distance to Ohio injection wells. Small firms complete new wells less often and therefore experience greater month-to-month variability in their capacity to reuse wastewater internally.

Sharing may be easier when firms use similar fracking fluids and produce wastewater of similar chemical composition. To explore this possibility, Figure 3b and 3c depict two important geological features that potentially affect the optimal design of fracking fluid and wastewater composition: the thickness of the Marcellus shale (Figure 3b) and the location of the Rome Trough (Figure 3c). The former can directly influence the choice of proppant (sand) and chemical additives, while the latter predicts historical temperature and pressure exposures affecting hydrocarbon characteristics. For comparison, Figure 3e shows the percentage of wastewater shipped to rival well pads or CTFs by well pad. Sharing appears to be less common in the far northeast where the Marcellus Shale is particularly thick, and more common within the broad region underlying the Rome Trough.

Because it is costly to transport wastewater, distance appears to be an especially important factor in shaping matching patterns. For more than half of all shipments, the destination facility was one of the five nearest facilities appearing as a destination for wastewater disposal from any origin well pad in the same month. The last section of Table 2 shows the distribution of shipment distance by destination type. The mean shipment distance was 30.0 miles. In comparison, the average distance between a randomly selected well pad-destination pair is roughly five times greater. Shipments to rival well pads and CTFs tend to be slightly longer than shipments to internal well pads or CTFs, but shipments to injection disposal wells originating from the same locations would typically be much longer.

2.2.2 Evidence of transaction costs

In November 2017 EQT Corporation (“EQT”) and Rice Energy Inc. (“Rice”) merged, creating the largest natural gas producer in the United States. Prior to merging, EQT and Rice both

re-transferred to rival firms, which is highly implausible.

operated large numbers of wells on overlapping acreage in far southwestern Pennsylvania.²²

The locations of EQT and Rice well pads prior to the merger are depicted in Figure 4. In the six months leading up to the June 2017 merger announcement, 98% of Rice’s wastewater disposal originated from well pads within 20 miles of an EQT facility that appeared as a destination for wastewater disposal in the same month. 64% of EQT’s wastewater disposal originated at well pads within 20 miles miles of a Rice facility that appeared as a destination. Nevertheless, EQT and Rice did not share prior to the merger, although EQT shared with firms other than Rice.²³ This can be seen in Table 4, which describes wastewater disposal from former EQT and Rice well pads before and after the merger.

Table 4 also shows that shipments between formerly-unintegrated EQT/Rice well pads increased dramatically after the merger. Between November 2017 and the end of the sample, 22.5% of wastewater generated at former EQT well pads was transferred to former Rice well pads, while 62.4% of wastewater generated at former Rice well pads was transferred to former EQT well pads. Hence, the removal of the firm boundary was followed by an immediate and significant increase in “sharing.” It is important to recognize that this outcome need not have occurred. For example, if the main factor preventing sharing in the period prior to the merger were technological in nature, the merger need not have been followed by any “sharing” at all. In Coase (1937), transaction costs are eliminated by integration. Thus, I interpret these changes as evidence that the merger eliminated Coasean transaction costs.

2.3 Additional comments

Here I briefly remark on two limitations of the data. Further details are provided in Appendix A.1. First, the data reports wastewater transfers rather than wastewater generation. If wastewater is re-used without leaving a well pad, it is not necessarily recorded in the data. Consequently, the total volume of wastewater disposals originating from a given well pad in a given month as observed in the data is potentially endogenous. In the analysis below, I abstract from this possibility. This choice is reasonable provided that on-site storage capacity is small relative to wastewater production for a typical well.

In the case that wastewater is reused, the data does not indicate what treatments were performed or where treatment occurred. This information is not essential for the empirical strategy on account of the transferrable utility assumption.

²²EQT was also present in West Virginia and northeastern Pennsylvania; Rice had a presence in Ohio.

²³Rice did not send wastewater to any rival firms in the pre-merger period. Rice did receive a small amount of wastewater from well pads associated with Alpha Shale Resources. However, Rice and Alpha Shale Resources had formerly been partners in a joint venture.

3 Model

This section presents the model. The model describes how firms minimize water-related costs, including the choice between internal reuse, sharing, and injection disposal.

Let K_t denote the finite set of well pads generating wastewater in month t , and D_t the finite set of facilities that are able to accept shipments of wastewater. For now, I assume D_t includes two types of facilities: well pads that are currently capable of accepting wastewater for reuse, and injection wells that conduct final disposal. (CTFs are discussed below.) Well pad $\kappa \in K_t$ is exogenously endowed with $Q_{t\kappa}$ truckloads of unstoreable wastewater that must be transported immediately to some facility in D_t . The capacity of facility $\delta \in D_t$ is exogenous and denoted by $C_{t\delta}$. For the remainder of this section I suppress the index t .

To describe the equilibrium, I denote the set of well pads currently capable of accepting wastewater for reuse by D^1 and the set of injection wells by D^0 . The cost of transporting a truckload of wastewater i from a well pad $\kappa \in K$ to another well pad $\delta \in D^1$ to satisfying the j th incremental unit of demand for source water at δ is given by

$$r_{\kappa\delta} - \epsilon_{i\delta} - \eta_{\kappa j}$$

where $r_{\kappa\delta}$ denotes the systematic cost of reuse and $\epsilon_{i\delta}$ and $\eta_{\kappa j}$ capture shipment-specific cost shocks incurred at the origin and destination facilities, respectively. The systematic cost of reuse may include (variable) transportation costs, transaction costs (in the case of sharing), and other facility-pair characteristics potentially affecting the cost of reuse.

By shipping truckload i from $\kappa \in K$ to a particular well pad $\delta \in D^1$, the firm operating κ incurs a cost $r_{\kappa\delta}^K - \epsilon_{i\delta}$. By accepting truckload j specifically from κ , the firm operating δ incurs a cost $r_{\kappa\delta}^D - \eta_{\kappa j}$. Utility is transferable in the sense that $r_{\kappa\delta}^K + r_{\kappa\delta}^D = r_{\kappa\delta}$.

For the firm operating well pad $\kappa \in K$, the alternative to reuse is injection disposal. Injection well owners are assumed to accept shipments on a posted price basis without consideration for the characteristics of well pad κ (such as operator identity or wastewater quality). The cost of injection disposal at $\delta \in D^0$ for truckload i from κ is given by $r_{\kappa\delta}^{inj} + p_\delta - \epsilon_{i\delta}$, where $r_{\kappa\delta}^{inj}$ denotes the (sender's) systematic cost and p_δ denotes a facility-specific posted price. For the firm operating well pad $\delta \in D^1$, the alternative to reuse is freshwater. The cost of sourcing one truckload of freshwater j for use at well pad δ is $r_{0\delta}^{fresh} - \eta_{0j}$.

A key assumption in any matching model concerns the distribution of cost shocks $\epsilon_i = (\epsilon_{i0}, \dots, \epsilon_{iD})$ and $\eta_j = (\eta_{0j}, \dots, \eta_{Kj})$. Point identification of the systematic costs is contingent on the assumption that ϵ_i and η_j are additively separable, mutually independent, and drawn from known distributions (Galichon and Salanié, 2022; Gualdani and Sinha, 2023). For the sake of tractability, I assume that ϵ_i and η_j consist of mutually independent draws

from extreme value type I distributions with scale parameters σ_ϵ and σ_η . As explained below, this assumption greatly simplifies estimation and counterfactual simulation.²⁴

The equilibrium concept is stable matching. A *feasible matching* assigns each truckload of wastewater from each well pad $\kappa \in K$ to some facility $\delta \in D$. In this setting, a feasible matching is *stable* if no firm (or pair of firms) can unilaterally (or jointly) reduce costs by swapping the destinations of two truckloads of wastewater i and i' , and no firm assigned to receive wastewater can unilaterally reduce costs by sourcing more freshwater instead.

Suppose the total volume of wastewater generated and accepted in a month grows uncountably large in proportion to $\mathbf{Q} = (Q_1, \dots, Q_K)$ and $\mathbf{C} = (C_1, \dots, C_D)$. Under mild regularity conditions, there exists in this “large market” limit a unique stable matching in which truckload i from $\kappa \in K$ is shipped to the least-cost destination

$$\min_{\delta \in D} r_{\kappa\delta}^K - \epsilon_{i\delta}. \quad (1)$$

where $r_{\kappa\delta}^K = r_{\kappa\delta}^{inj} + p_\delta$ if $\delta \in D^0$. Simultaneously, each incremental unit of capacity j at each well pad $\delta \in D^1$ is allocated to the least-cost source of base fluid

$$\min_{\kappa \in \{0\} \cup K} r_{\kappa\delta}^D - \eta_{\kappa j}. \quad (2)$$

where $r_{0\delta}^D = r_{0\delta}^{fresh}$. In the stable match, senders’ demand to ship wastewater from κ to δ (according to (1)) exactly coincides with receivers’ demand for wastewater from κ at δ (according to (2), see Gretsky et al., 1992; Galichon and Salanié, 2022).

A matching can be described by a non-negative $|K| \times |D|$ matrix μ in which element $\mu_{\kappa\delta}$ represents the mass of shipments between κ and δ resulting from the match. Shipments of freshwater to $\delta \in D^1$ are denoted by $\mu_{0\delta}$. With extreme value type I cost shocks ϵ_i and η_j , the unique stable matching μ^* satisfies

$$\mu_{\kappa\delta}^* = \begin{cases} Q_\kappa^\gamma C_\delta^{1-\gamma} \exp \{ (\sigma_\epsilon + \sigma_\eta)^{-1} r_{\kappa\delta} + \gamma F_\kappa + (1 - \gamma) H_\delta \} & \text{if } \kappa\delta \in K \times D^1 \\ Q_\kappa \exp \{ \sigma_\epsilon^{-1} (r_{\kappa\delta}^{inj} + p_\delta) + F_\kappa \} & \text{if } \kappa\delta \in K \times D^0 \\ C_\delta \exp \{ \sigma_\eta^{-1} r_{\kappa\delta}^{fresh} + H_\delta \} & \text{if } \kappa\delta \in \{0\} \times D^1 \end{cases} \quad (3)$$

where (F, H, p) are determined in equilibrium. The first case encompasses shipments for reuse. F_κ is the inclusive value associated with (1), H_δ is the inclusive value associated with (2), and $\gamma = \sigma_\epsilon^{-1} / (\sigma_\epsilon^{-1} + \sigma_\eta^{-1})$ captures the relative dispersion of ϵ_i and η_j . The second and

²⁴As in discrete choice models, logit errors severely restrict substitution. Galichon and Salanié (2022) propose a nested logit and other extensions; Gualdani and Sinha (2023) a fully nonparametric approach.

third cases describe shipments to injection wells and shipments of freshwater, respectively.

Given $(\sigma_\epsilon, \sigma_\eta, r, Q, C)$, the $|K| + |D|$ -dimensional vector (F, H, p) is pinned down by the following $|K| + |D|$ market clearing conditions:

$$\sum_{\delta \in D^0} \mu_{\kappa\delta}^* + \sum_{\delta \in D^1} \mu_{\kappa\delta}^* = Q_\kappa \quad \forall \kappa \in K^1 \quad (4)$$

$$\mu_{0\delta}^* + \sum_{\kappa \in K} \mu_{\kappa\delta}^* = C_\delta \quad \forall \delta \in D^1 \quad (5)$$

$$\sum_{\kappa \in K} \mu_{\kappa\delta}^* = C_\delta \quad \forall \delta \in D^0 \quad (6)$$

If $\sigma_\epsilon = \sigma_\eta$ (as maintained by Choo and Siow, 2006), (4)-(6) constitute a biquadratic system of equations. Galichon and Salanié (2022) propose a fast procedure based on Sinkhorn’s algorithm for this special case, which can easily be adapted here. In the interest of computational efficiency I therefore assume that $\sigma_\epsilon = \sigma_\eta$ below.

Centralized treatment facilities (CTFs) When taking the model to data, I include third party CTFs in D^0 and operator-affiliated CTFs in D^1 . Thus, third party CTFs are assumed to indiscriminately accept wastewater from all sources at posted prices, while oil and gas firms operating their own CTFs are assumed to be selective in deciding where to source wastewater and may opt for freshwater in downstream operations. These assumptions may be imperfect. Third party CTFs generally operate on a zero liquid discharge principle; in equilibrium, each barrel of wastewater accepted by a CTF therefore substitutes for freshwater in some later fracking operation. Of greater concern is the potential for endogenous storage behavior. While well pads are space constrained and therefore tend to accept wastewater just-in-time to complete fracking operations, CTFs can instead accept wastewater for the purpose of future reuse. I do not attempt to incorporate dynamic incentives because I do not observe monthly CTF inventory levels, though some prior work on marriage markets has extended the basic TU matching model in this direction (e.g., Choo, 2015).

Interpretation The model assumes that firms make decisions on a truckload-by-truckload basis, both when sending and receiving wastewater. This assumption yields a convenient model of the internal operations of the firm. It is a reasonable assumption under the assumption that firms are price-takers: the firm can be viewed as a coalition of individual managers; in the core (here, the stable matching), no coalition of managers can achieve lower costs than managers acting independently (Shapley and Shubik, 1971). The model implies that transactions within the firm are “priced,” but these prices have a natural interpretation

in terms of the shadow price of scarce capacity. If the capacity to reuse wastewater at a particular well pad is limited (by engineering specifications, for example), accepting wastewater transported from one facility precludes the acceptance of wastewater from another.

Limitations An important limitation of the model is that firm-level strategic behavior is precluded. In industrial organization, this type of behavior is often modeled using Horn and Wolinsky (1988)’s “Nash-in-Nash” bargaining model. Like the model in this paper, Nash-in-Nash presumes that firms jointly maximize surplus within relationships (here, conditional on transaction costs).²⁵ However, relationships are formed strategically. One challenge in adapting the Nash-in-Nash framework to the setting in this paper is the difficulty of constructing counterfactuals in which the network of sharing relationships is allowed to adjust.²⁶ As the results will show, policies which facilitate relationship formation are potentially a promising complement to conventional Pigouvian regulation of injection disposal.

Another important restriction imposed by the model is that systematic costs are incurred in proportion to wastewater volume. In fact, many examples of transaction costs are best understood as fixed costs that might be incurred once for multiple shipments (e.g., search costs), or even once per relationship (e.g., the cost of drafting a first contract). In the model, such costs are rationalized by higher variable transaction costs. Explicitly accounting for relationship-specific fixed transaction costs would require a considerably more complex model, as for example in models of trade with multinational firms (e.g., Tintelnot, 2017).

4 Estimation

This section describes the estimation procedure. I derive a Poisson pseudomaximum likelihood estimator via the method of moments (Galichon and Salanié, 2022, 2024).

Parameterization Because freshwater acquisition costs in Pennsylvania are modest and tend to exhibit less variation than injection disposal costs, I impose $r_{0\delta}^{fresh} = 0$ for all δ . The key objects to be estimated are therefore the systematic costs of reuse and injection disposal r . Among the shipments for reuse $\kappa\delta \in K \times D^1$, I specify

$$\{\sigma_\epsilon + \sigma_\eta\}^{-1} r_{\kappa\delta} = x'_{\kappa\delta} \theta. \quad (7)$$

²⁵It is interesting to note that models of Nash-in-Nash bargaining typically do not account for transaction costs, either within relationships or in the process of relation formation (Carlton, 2020).

²⁶Ho and Lee (2019) and Ghili (2022) develop models of network formation in Nash-in-Nash environments. These papers simplify the strategy space by exploiting institutional differences between upstream and downstream firms that have no clear analogues in the wastewater reuse setting.

For the primary specifications of interest, the covariate vector $x_{\kappa\delta} = (x_{\kappa\delta}^{(1)}, \dots, x_{\kappa\delta}^{(J)})$ includes distance $x_{\kappa\delta}^{(1)}$ and dummy variables $x_{\kappa\delta}^{(1)}, \dots, x_{\kappa\delta}^{(J)}$ describing the relationship between the operating firms at κ and δ . I also consider interactions of geological characteristics affecting well productivity and frack design. It is generally possible to accommodate high dimensional fixed effects using techniques from the literature on gravity models, as explained below.

For injection disposal (and third party CTFs) $\kappa\delta \in K \times D^0$, I specify

$$(1 - \gamma) \sigma_\epsilon^{-1} r_{\kappa\delta}^{inj} = x'_{\kappa\delta} \theta. \quad (8)$$

The normalization in (8) differs from (7) in order to ensure that any costs common to injection disposal and reuse (such as trucking costs) are invariant to whether the destination δ is a well pad or injection well. For the primary specifications of interest, $x_{\kappa\delta}$ will include only the distance from κ to δ in the case of injection disposal.

In reality, firms might incur Coasean transaction costs not only when sharing wastewater with rival firms, but also when transferring wastewater to third party CTFs and injection wells (i.e., the facilities in D^0). Transaction costs in the second case are not separately identified from the prices p . For this reason the terms “firm boundary” and “transaction costs” below refer specifically to sharing transactions (as defined in Section 2.2). Transaction costs associated with third party firms are likely to be much smaller than transaction costs among rival oil and gas producers, for two reasons. First, rival oil and gas firms compete in a variety of upstream and downstream markets (leasing, labor, pipeline capacity, etc.), which likely discourages cooperation. Second, oil and gas firms may have fewer resources for coordinating the receipt of another firm’s wastewater than specialized disposal firms.

Moment-matching procedure Let $y_{\kappa\delta}$ denote the observed quantity of wastewater shipments from κ to δ in month t . I again omit time subscripts, treating each observed month of data as a separate matching market for the purpose of estimation.

Following Galichon and Salanié (2022), I assume that the facility capacities $\hat{Q}_\kappa = \sum_{\delta \in D} y_{\kappa\delta}$ and $\hat{C}_\delta = \sum_{\kappa \in K} y_{\kappa\delta}$ are observed without sampling error. Let $\mu^\theta(Q, C)$ denote the true but unobserved matching given θ and capacity vectors $Q := (Q_1, \dots, Q_{|K|})$ and $C := (C_1, \dots, C_{|D^1|})$ and recall that $x_{\kappa\delta} = (x_{\kappa\delta}^{(1)}, \dots, x_{\kappa\delta}^{(J)})$. For each $j = 1, \dots, J$, I impose that:

$$E \left[\left(y_{\kappa\delta} - \mu_{\kappa\delta}^\theta \left(\hat{Q}, \hat{C} \right) \right) x_{\kappa\delta}^{(j)} \right] = 0 \quad \forall \kappa\delta \in K \times D \quad (9)$$

The empirical analogue to the left hand side of (9) is

$$g_j(\theta) = \sum_{(\kappa, \delta) \in K \times D^1} (y_{\kappa\delta} - \mu_{\kappa\delta}^\theta) x_{\kappa\delta}^{(j)}, \quad (10)$$

The method of moments estimator $\hat{\theta}$ is a root of $g(\theta) := (g_1(\theta), \dots, g_J(\theta)) = 0$.

The moment condition (9) requires that the difference between observed and predicted shipments is uncorrelated with observable facility-pair characteristics. This assumption may be violated if there are omitted factors that shift the systematic cost of reuse for both the sending and receiving firm (as opposed to their separable, idiosyncratic shocks ϵ and η).²⁷ On the other hand, facility-specific omitted factors that shift the cost of reuse at a particular location $\delta \in D_1$ for all sources of wastewater K are absorbed in the facility fixed effect H_δ .

The use of the method of moments rather than the generalized method of moments (GMM) is motivated by the observation that the empirical moment conditions together with the empirical analogues of (4)-(6) together coincide with the first order conditions of the following Poisson pseudomaximum likelihood (PPML) estimator:

$$\begin{aligned} \max_{\theta, F, H, p} \quad & \sum_{\kappa\delta \in K \times D^1} \{y_{\kappa\delta} \log \mu_{\kappa\delta}^\theta - \mu_{\kappa\delta}^\theta\} \\ & + \sum_{\kappa\delta \in K \times D^0} \gamma \{y_{\kappa\delta} \log \mu_{\kappa\delta}^\theta - \mu_{\kappa\delta}^\theta\} + (1 - \gamma) \{y_{0\delta} \log \mu_{0\delta}^\theta - \mu_{0\delta}^\theta\} \end{aligned} \quad (11)$$

PPML is appealing here because the wastewater disposition data resembles typical trade datasets in which errors are heteroskedastic and the dependent variable frequently takes a value of zero. PPML is widely used in the literature on trade due to its favorable performance in such datasets and its relative simplicity (Santos Silva and Tenreyro, 2022).

One difference between (11) and the standard gravity model is that shipments involving the outside options (injection disposal, freshwater) appear with the weights γ and $1 - \gamma$. The sending-facility fixed effect in $\log \mu_{\kappa\delta}^\theta$ is γF_κ for $\delta \in D^1$ but F_κ for $\delta \in D^0$. The model is therefore linear only conditional on γ . For this reason, it is convenient to fix γ . As mentioned above, I follow the literature and impose $\sigma_\epsilon = \sigma_\eta$ which implies $\gamma = 0.5$.²⁸ Conditional on γ , (11) is computationally feasible despite the large dimension of (F, H, p) : as in the estimation of gravity models, it can be solved quickly using iteratively reweighted least squares (IRLS).²⁹

²⁷For an instructive discussion of separability see Chiappori (2017). In a recent contribution, Adda et al. (2025) propose a model which relaxes separability with marriage-specific Gaussian shocks.

²⁸In principle γ is identified from the relative rates at which firms substitute to the outside options as the sizes of K_t and D_t vary. However, estimation of γ significantly complicates inference.

²⁹IRLS is the typical method for estimating Poisson regression models with high-dimensional fixed effects. The Frisch-Waugh-Lovell theorem is exploited to avoid inverting the full design matrix (Correia et al., 2020).

Inference In the context of gravity models, Fernandez-Val and Weidner (2016) demonstrate that estimation of θ is not affected by the incidental parameters problem even when the number of fixed effects is large. However, the incidental parameters problem does contribute to biased standard errors and incorrectly centered confidence intervals for inference based on the score of the pseudolikelihood (Kauermann and Carroll, 2001). Zylkin (2024) proposes to address this issue using a two-step bootstrap. I adopt the same procedure.

Identification Identification of the model is formally established in Galichon and Salanié (2022). Informally, there are two key objects. The first is the sum of the dispersions $\sigma_\epsilon + \sigma_\eta$. As in discrete choice models, a normalization is necessary. For this purpose I normalize the cost of distance to unity. For $\sigma_\epsilon + \sigma_\eta \rightarrow 0$, μ^θ converges to the solution to a linear program in which the objective depends only on the systematic costs r . For $\sigma_\epsilon + \sigma_\eta \rightarrow \infty$, μ^θ converges to random matching. $\sigma_\epsilon + \sigma_\eta$ is therefore identified by the amount of randomness in the equilibrium matching conditional on r that is necessary to satisfy the moment conditions. The second key object is the transaction costs. The transaction cost incurred when Firm A transfers wastewater to Firm B corresponds to the reduction in distance at which Firm A would be indifferent between transferring wastewater to Firm B versus its own facilities (or third party facilities). In the data, this requires that we observe Firm A shipping wastewater to locations that are sufficiently further away than Firm B’s facilities.

Freshwater consumption Implementation of (11) requires knowledge of $y_{0\delta}$, the amount of unmatched reuse capacity at facility δ in month t met with freshwater. This information is not contained in the disposal data, but can be estimated from FracFocus disclosures.³⁰ The details of the estimation procedure I adopt for this purpose are described in Appendix A.2. This procedure requires additional assumptions and introduces sampling error, potentially contaminating the estimates of θ . To mitigate this I adopt a two step approach to estimation. First, I drop the freshwater margin from (11) and estimate

$$\max_{\theta, F, H, p} \sum_{\kappa\delta \in K \times D^1} \{y_{\kappa\delta} \log \tilde{\mu}_{\kappa\delta}^\theta - \tilde{\mu}_{\kappa\delta}^\theta\} + \sum_{\kappa\delta \in K \times D^0} \gamma \{y_{\kappa\delta} \log \tilde{\mu}_{\kappa\delta}^\theta - \tilde{\mu}_{\kappa\delta}^\theta\} \quad (12)$$

with $\tilde{\mu}_{\kappa\delta}^\theta = \mu_{\kappa\delta}^\theta(\hat{Q}, \tilde{C})$ where $\tilde{C}_\delta$ is equal to $\sum_{\kappa \in K} y_{\kappa\delta}$ if $\delta \in D^1$ and $\hat{C}_\delta$ otherwise. This estimator is sufficient to recover θ and hence most parameters of interest, including the transaction costs of sharing. However, the resulting estimates are not sufficient for counterfactual analysis because the vector (F, H, p) is only identified up to location in this case. In the

³⁰FracFocus is a public registry in which firms disclose chemicals used in fracking particular wells (including water). In some states, disclosure is voluntary. In Pennsylvania, it is mandatory for unconventional wells.

second step, I recover the location of (F, H, p) , which I denote by ζ . Given $r_{0\delta}^{fresh} = 0$, (2) implies that freshwater consumption as a function of ζ is given by

$$\mu_{0\delta}^{\theta}(\zeta) = \frac{\tilde{C}_{\delta}}{1 + \tilde{C}_{\delta} \exp \left\{ - (1 - \gamma)^{-1} \left(\hat{H}_{\delta}^0 - \zeta \right) \right\}}$$

where $\hat{H}^0$ is the estimate of H obtained from (12). I estimate ζ by matching model-implied freshwater consumption to the freshwater consumption estimates $\hat{y}_{0\delta}$:

$$\hat{\zeta} = \arg \min_{\zeta} \left(\sum_{\delta \in D^1} \hat{y}_{0\delta} - \sum_{\delta \in D^1} \mu_{0\delta}^{\theta}(\zeta) \right)^2$$

This approach allows me to construct counterfactuals in which shipment elasticities and transaction costs are estimated separately from the (possibly noisy) freshwater estimates.

Self-shipments 8.7% of wastewater disposal in the data originates at a well pad that is observed to receive wastewater in the same month: that is, for a pair of indices κ^* and δ^* corresponding to the same underlying facility, I observe both $\sum_{\delta \in D} y_{\kappa^* \delta} > 0$ and $\sum_{\kappa \in K} y_{\kappa \delta^*} > 0$. Although it is never the case that $y_{\kappa^* \delta^*} > 0$ empirically, the model can predict $\mu_{\kappa^* \delta^*} > 0$ in this case.³¹ To exclude this possibility I impose $r_{\kappa^* \delta^*} = \infty$. This assumption also serves the purpose of eliminating physically infeasible shipments in the case that $\sum_{\delta \in D} y_{\kappa^* \delta}$ includes wastewater generated by wells that were completed using $\sum_{\kappa \in K} y_{\kappa \delta^*}$.

Separation For some results, I entertain specifications in which $x_{\kappa\delta}$ contains many fixed effects. For example, I estimate models with firm-pair specific transaction costs. A general problem in PPML estimation (and other applications of maximum likelihood estimation) is that finite point estimates may fail to exist for certain parameters, particularly in the presence of high-dimensional fixed effects. For example, if there is no sharing within a given firm pair, the PPML estimate of the associated firm-pair specific transaction cost will diverge to $+\infty$. This problem is more generally known as the separation problem. With overlapping fixed effects, such as firm pair dummies combined with interactions between well characteristics, separation can manifest in subtle ways. To address it, I adopt an algorithm from Correia et al. (2021) to identify any separated “cells.” Because $y_{\kappa\delta} = 0$ for affected facility pairs and because $y_{\kappa\delta} \log \mu_{\kappa\delta}^{\theta} \rightarrow 0$ as $\mu_{\kappa\delta}^{\theta} \rightarrow 0$, I adopt the convention that $y_{\kappa\delta} \log \mu_{\kappa\delta}^{\theta} = 0$ and $\mu_{\kappa\delta}^{\theta} = 0$ when separation occurs. This approach is standard in statistical packages designed for gravity estimation (such as Correia et al., 2020).

³¹Indeed, this will necessarily be the case for finite $r_{\kappa^* \delta^*}$ on account of the support of the logit errors.

Trucking costs Converting point estimates such as those in Table 5 into dollars requires an estimate of the marginal cost of trucking. There is limited public information on the cost of wastewater hauling, which is typically intermediated at arms length between an operating firm and a wastewater trucking company. My preferred estimate is \$69 dollars per hour, or \$2.50 per mile. This estimate is obtained by adding together estimates of driver wages and fuel costs and then adjusting for empty (“deadhead”) return trips (see Appendix A.3). The estimate is intended to be conservative: I do not account for vehicle depreciation, maintenance, overtime wages, or markups. Trucking cost estimates in the literature range as high as \$175/hour and \$4.3/mile (Puder and Veil, 2006; Tavakkoli et al., 2020).

5 Estimates

This section describes the model estimates. I first explain how the parameter estimates are interpreted in the context of a series of preliminary specifications. This exercise also serves to validate the model. Then I discuss the transaction cost estimates in detail, focusing on the extent of variation across relationships. Finally, I discuss producer surplus and its implications for the overall significance of reuse and sharing in Pennsylvania.

5.1 Preliminary estimates

Table 5 presents parameter estimates for a series of preliminary models. Each model includes a measure of distance. In Columns (1) to (2), this is the log of over-the-road mileage between well pad κ and destination δ (“Log Miles”). I focus first on Columns (1) and (2). In Column (1), distance is the only explanatory variable (apart from the implicit origin-month and destination-month fixed effects F , H , and p). The coefficient on distance implies that the elasticity of shipment with respect to distance is -1.705 . Column (2) includes a dummy for whether the origin and destination facilities are operated by different firms (i.e., whether a firm boundary is present).³² In this case, the elasticity of shipment with respect to distance is -1.062 . The second estimate is similar to typical distance elasticities in the trade literature, which are often close to -1 .³³ Model fit in Column (2) is improved relative to Column (1) in terms of the squared correlation between predicted and observed shipments (reported as R^2) and deviance. Thus, firm boundaries appear to be important for explaining reuse.

³²This variable captures sharing transactions. As explained above, I do not account for firm boundaries in the case of shipments to third party CTFs or injection wells.

³³Head and Meyer (2014) report -1.1 to be the mean distance elasticity of trade across a large number of studies estimating structural gravity models. Chaney (2018) investigates this regularity. In their sample, Atalay et al. (2019) obtain an estimate of -0.96 .

The models in Columns (3) to (6) replace log mileage with over-the-road mileage itself (“Miles”). Comparing (2) and (4), linear distance improves model fit when accounting for firm boundaries. While log distance affords a convenient interpretation, linear distance may be more suitable in the wastewater management context if trucking contracts feature explicit linear elements (such as a cost per-mile or cost per-hour). The improvement in fit therefore motivates me to use linear rather than log distance for the remainder of the paper.

The estimates in Column (4) imply that crossing a firm boundary (in other words, sharing) increases the cost of reuse by an amount equivalent to the cost of shipping a truckload of wastewater $\frac{4.958}{0.036} = 137.7$ additional miles, nearly five times the mean observed shipment distance. This statistic can be viewed as a coarse measure of average transaction costs. If the cost of trucking is \$2.50 per mile (see above), it implies that crossing a firm boundary costs roughly \$3.50 per barrel of wastewater. This cost is larger than the typical treatment costs and disposal fees discussed in Section 2.

I assume throughout the paper that there are no transaction costs within firms. To test this assumption, Columns (2) and (4) also include an indicator for post-merger transactions across the pre-merger firm boundary between EQT and Rice. In Column (2), the coefficient on this indicator is positive but insignificant at the 5% level. In Column (4), the coefficient is negative and statistically significant but small. A small and negative estimate implies a small and positive cost of crossing the pre-merger firm boundary. In Column (4) this cost is estimated to be 3.7% as large as the cost of sharing: although not a statistical zero, the effect is economically small. I view this result as broadly consistent with a Coasean interpretation of the estimated firm boundary effects: sharing transactions are subject to significant frictions that do not affect shipments for reuse within the firm.³⁴

Although the EQT/Rice merger evidence in Section 2.2 clearly indicates that some pairs of firms do not engage in sharing despite apparent gains from trade, one might hypothesize that trade remains efficient within established “sharing relationships.” To test this hypothesis, Column (5) interacts the firm boundary dummy with an indicator for whether any shipments are observed from the sender’s well pads to the receiver’s facilities in the same month. When this is the case, I say that firms have an active relationship. Observations corresponding to inactive relationships are necessarily separated, causing the large decrease in the effective sample size. Within active relationships, the firm boundary effect is statistically significant at the 5% level and equivalent to the cost of shipping a truckload of wastewater $\frac{2.342}{0.034} = 68.9$ additional miles, or about \$1.50 per barrel of wastewater. Hence, the firm boundary effect remains economically large on the intensive margin.

Column (6) augments the specification in Columns (4) with controls for geological char-

³⁴Residual within-firm friction is not surprising if post-merger integration is not completed immediately.

acteristics of the Marcellus shale beneath the origin and destination facilities. These controls are meant to address the possibility that apparent firm boundary effects reflect engineering frictions related to wastewater composition or fracking fluid design. If apparent firm boundary effects are attributable to heterogeneity in well types across operators, one would expect the standard error of the firm boundary coefficient to increase significantly in Column (6). This does not occur, and model fit does not substantially improve. Firm boundaries thus appear to play a more important role in explaining reuse than well characteristics.

5.2 Transaction cost heterogeneity

In this section, I estimate more flexible parameterizations of transaction costs in order to examine how the effects of firm boundaries differ across relationships and over time.

Analysis of variance Table 6 reports model fit statistics from a sequence of models in which firm boundary effects are allowed to vary with transaction characteristics. Rows 1. and 2. correspond to columns (2) and (4) of Table 5, respectively.³⁵ I report the deviance of the estimated model based only on potential shipments for reuse (i.e., from K to D^1) along with two separate pseudo- R^2 statistics.³⁶ R_1^2 captures the improvement in model fit relative to the specification in row 1 (i.e., a model without any firm boundaries), while R_2^2 captures the improvement in model fit relative to the specification in row 2 (i.e., a model with constant firm boundary effects).

Row 4. shows that interacting the firm boundary dummy with firm identity dummies generates considerable improvement versus row 2. ($R_2^2 = 0.093$). This indicates that some firms experience higher transaction costs than others. Row 5. shows that there is further improvement when effects are allowed to differ depending on whether a firm is a sender or receiver ($R_2^2 = 0.123$). Hence, the transaction costs a firm encounters when sending wastewater to rivals are generally different from those encountered on receipt. Row 7. shows that sending firm and receiving firm transaction costs can differ substantially by counterparty ($R_2^2 = 0.214$). This implies that transaction costs are generally relationship-specific. Moreover, the difference between rows 6. and 7. implies that transaction costs can be asymmetric within relationships ($R_2^2 = 0.180$ when transaction costs are assumed to be symmetric within relationships versus $R_2^2 = 0.214$ when they are not). Taken together, these findings suggest that sharing is mediated by an incomplete network of frictional bilateral relationships, where

³⁵For this exercise, I drop the EQT-Rice Post-Merger dummy and the geologic controls.

³⁶The focus on shipments for reuse emphasizes the impact of transaction costs that affect sharing. I obtain similar results when also including shipments for injection disposal in the deviance calculation.

relationships are frictional in the sense that transaction costs can exist within them.³⁷

Row 3. shows that interacting the firm boundary dummy with time (month) dummies generates little very little improvement in model fit ($R_{base}^2 = 0.007$). Hence, the initial firm boundary estimate of \$3.50 per barrel of wastewater from Table 5 does not vary much over the course of the sample. This finding is reassuring that point estimates in Table 5 are robustly estimated. It also excludes a strong relationship between transaction costs and market conditions, to the extent that these changed during the sample period.³⁸ However, rows 9. and 10. indicate that transaction costs exhibit some change within relationships over time ($R_2^2 = 0.323$ versus $R_2^2 = 0.180$ in row 6. for the symmetric case, $R_2^2 = 0.337$ versus $R_2^2 = 0.214$ in row 7. for the asymmetric case). These findings can potentially be reconciled if firm-level variation in completion frequency (or location) drives month-to-month churn in sharing market participation while overall sharing activity remains flat.

Firm characteristics I next consider the extent to which the firm-level variation in transaction costs revealed in the previous section can be explained by observable firm characteristics. To build intuition, I begin by dividing firms into five distinct groups as in Section 2.1: Southwest Large, Southwest Small, Northeast Large, Northeast Small, and Peripheral. I then estimate a model allowing firm boundary effects to vary for each combination of sending and receiving firm type. The results are summarized in Table 7. The point estimates suggest that transaction costs are generally greater for shipments that cross regions, particularly when one or both firms is Small. This finding is consistent with a model in which relationships are more likely to form at shorter distances, when potential gains from trade may be larger (or search costs lower). In both regions, transaction costs are greater for shipments between Large firms than between Small firms. This differences is larger in the Northeast. If larger firms compete more directly in upstream or downstream markets, this finding could reflect a vertical foreclosure or raise rival’s cost mechanism that disincentives coordination. There is no clear relationship between relative firm size and transaction costs: in the Northeast transaction costs are lowest for shipments from Large firms to Small firms, while in the Southwest transaction cost are highest for such shipments.

Although these findings provide some limited evidence in support of specific mechanisms, firm size and location on the whole appear to explain little of the total variation in transaction costs across relationships. Rows 11. to 13. of Table 6 report model fit statistics for

³⁷This structure might be formalized in terms of a latent contract network with frictional trade (e.g., Fleiner et al., 2019). It is notable that in this connection that the ubiquity of incomplete, frictional trading networks in the real economy has motivated recent empirical work in international trade (e.g., Martin et al., 2023; Miyauchi, 2024) and development economics (e.g., Macchiavello and Morjaria, 2021).

³⁸From 2017 to 2020, the price of natural gas was low and essentially flat. I cannot rule out the possibility that the significance of firm boundaries might differ substantially under alternative market conditions.

specifications which allow for more flexibility in the effects of firm location and size than the model in in Table 7.³⁹ In comparison to the specifications in rows 4. through 10., neither firm size nor location substantially improves model fit relative to row 2. ($R^2_2 < 0.02$).

5.3 Producer surplus and output

As is well known, the welfare implications of oil and gas extraction are massive and wide-ranging. At the level of state policymaking, greater extraction comes with the promise of economic development and increased tax revenue but risks numerous local environmental harms. It is therefore important to understand the significance of reuse in firms’ drilling decisions before considering policy changes that might affect the cost of reuse.

To this end, I use the fitted model to quantify producer surplus gains from reuse and sharing under the status quo. For the remainder of the paper, the “baseline” model is the specification in row 10. of Table 6, which features time-varying, asymmetric, relationship-specific transaction costs. Using this specification, I predict that the average barrel of wastewater is shipped 93.1 miles fewer on account of reuse and sharing at current disposal costs.⁴⁰ This is reported in the final row of the Table 8. Evaluated at a conservative trucking cost estimate of \$2.50/mile, this implies shipping cost savings of \$2.12 per barrel of wastewater. On an output basis, savings of this magnitude amount to about \$0.020 per million British thermal unit (MMBtu) natural gas. This figure does not account for any distance-invariant efficiencies from reuse and sharing. Such efficiencies are likely modest: perhaps \$0.50-\$1.00 per barrel of wastewater based on the figures cited in Section 2.1. Total cost savings (before transaction costs) are therefore on the order of \$3.00 per barrel, or \$0.029 per MMBtu. This amounts to 1% the average spot price of natural gas at Henry Hub during the analysis period.⁴¹

Newell et al. (2019) estimate a supply elasticity of 0.89 for US gas production (including conventional production). Hausman and Kellogg (2015) estimate a long-run supply elasticity of 0.81 for shale gas. These estimates imply that cost savings on the order of 1% of the benchmark price of gas are consistent with an increase in marketed production of less than 1%. This suggests that most observed production in Pennsylvania would have occurred (at higher cost) even if reuse and sharing had not been possible, with marginal wells being concentrated in the Northeast.⁴² Even in the Northeast, Table 8 suggests that cost savings

³⁹For firm size, I use quartiles of the firm size distribution, as measured by gas production. For location, I use the USGS assessment unit containing the plurality of a firm’s wells.

⁴⁰To form this estimate, I compare the quantity-weighted average shipment distance in the estimated model with the quantity-weighted average shipment distance if reuse was not possible. To calculate the latter, I ignore injection well capacity constraints and hold fixed the estimated injection well prices p .

⁴¹Equal to \$2.69 per MMBtu based on monthly average spot price from EIA.

⁴²Put differently, past improvements in water efficiency due to reuse and sharing in Pennsylvania are unlikely to have “backfired” in the sense of causing more drilling to occur.

are no more than 2% of the benchmark price of gas. Given that injection disposal costs in the Northeast are extreme by industry standards, this suggests that future policymaking related to water and wastewater is unlikely to have a large direct effect on extraction.

6 Equilibrium impacts of disposal fees

In this section, I use the estimated model to examine the impact of Ohio’s existing injection disposal fee, as well as proposed policy reforms.

Policy history In 2010, the Ohio legislature instituted \$0.05 and \$0.20 per barrel fees on “in-district” and “out-of-district” brine disposal, respectively, with the latter covering all out-of-state disposal (mainly from Pennsylvania and West Virginia). Importantly, fees are assessed only the first 500 Mbbl per well per year. Hence, higher-throughput wells face a lower effective fee. Ohio policymakers have subsequently sought to raise the disposal fee on two occasions. In 2012, Ohio Governor John Kasich unsuccessfully proposed to raise the out-of-district fee to \$1.00 barrel. In 2023, the Ohio House of Representatives passed budget legislation that would have “flattened” existing fees by removing the 500 Mbbl threshold. This provision was ultimately eliminated after reconciliation with an Ohio Senate bill. Thus, the original fee structure remains in place. Ohio residents have continued to express concerns over the volume of injection disposal occurring in the state (Sour, 2025).

Counterfactual strategy I focus on three policy counterfactuals: elimination of the current fees, removal of the 500 Mbbl threshold, and a flat \$1 per barrel fee. To construct these counterfactuals, I must address two key challenges. First, I do not observe pass through of the existing out-of-district fee to Pennsylvania oil and gas firms. Second, I must adjust the model to allow the volume injection disposal to change with the disposal fees.

To address the first problem, I assume that injection well operators pass through 100% of average out-of-district disposal fees to out-of-district disposal customers. This assumption is reasonable if the injection well sector is competitive and if marginal costs in the injection well sector are constant. To address the second problem, I move from a balanced to an unbalanced matching framework and adjust the cost of firms’ outside options. This approach allows me to predict the impact of counterfactual disposal fees in a localized sense without introducing a formal model of capacity investment in the injection well sector.

The model presented in Section 3 can be described as a balanced matching model: for each well pad $\delta \in D^1$, the observed level of reuse $\tilde{C}_\delta$ at δ is held fixed in every feasible match.

In an unbalanced matching model, by contrast, the level of reuse at δ is endogenous. This feature is necessary if the total amount of injection disposal is to change.

Beginning from the estimated model, I first aggregate all injection wells into a single “outside option.” In particular, I assume that the cost of the outside option for wastewater located at well pad $\kappa \in K$ is

$$V_{\kappa}^{inj} = \sigma_{\epsilon} \left(c + E \left[\max_{\delta \in D^0} r_{\kappa\delta}^{inj} + p_{\delta} + \epsilon_{i\delta} \right] \right) \quad (13)$$

where $c = 0.57721\dots$ is Euler’s constant. The predicted shipments for reuse $K \times D^1$ in an unbalanced matching model in which the cost of the outside option is given by (13) are identical to those implied by the fitted model. For the counterfactuals, I vary the cost of this outside option. In particular, I replace V_{κ}^{inj} with

$$V_{\kappa}^{cf} = \sigma_{\epsilon} \left(c + E \left[\max_{\delta \in D^0} r_{\kappa\delta}^{inj} + p_{\delta} + \Delta_{\delta}^{cf} + \epsilon_{i\delta} \right] \right) \quad (14)$$

where $\Delta_{\kappa\delta}^{cf}$ denotes a counterfactual change in the cost of disposal at injection well δ . The new matching is attained by solving the unbalanced matching problem described above with (14) as the cost of κ ’s outside option.

Computing $\Delta_{\kappa\delta}^{cf}$ entails two further challenges. First, even assuming 100% pass through, I do not directly observe average disposal fees at the well level on account of the 500 Mbbl threshold. Appendix A.4 describes the strategy I use to estimate average fees using injection well records from the Ohio DNR. Taking the resulting estimates as given, further assumptions are required to convert dollar values into units of utility as required by (14). This issue is addressed as follows. First, I use the \$2.50 per mile trucking cost estimate above to convert fees from dollars to miles. Second, I scale the distance-denominated fee using the distance coefficient from the estimated model. Under these assumptions, a \$0.20 per barrel fee has an equivalent cost impact on a Pennsylvania producer as moving all Ohio injection disposal wells 8.8 miles further away. A \$1.00 per barrel fee has an equivalent cost impact as moving all Ohio injection disposal wells 44.0 miles further away.

6.1 Results

Figure 5 plots the average injection disposal rate for Pennsylvania firms under the status quo and under the three counterfactuals described above. The dark blue line corresponds to the baseline specification of the model (row 10. in Table 6, reported here as Model 3). Without the existing \$0.20 per barrel fee, I estimate that injection disposal in Pennsylvania would

increase by 5.4 Mbbl/month (1.5% of the status quo). A “flattening” of the \$0.20 per barrel fee would cause a decrease of 2.7 Mbbl/month (0.7%). With a \$1.00 per barrel fee, injection disposal in Pennsylvania would decrease by 23.3 Mbbl/month (6.2%). Overall, these effects are quite modest, consistent with a relatively inelastic demand.

Flexibly accounting for transaction costs appears to be important for accurately estimating the above impacts. The red line in Figure 5 depicts estimates obtained with a model that does not account for transaction costs (row 1. in Table 6). In this case, the existing fee and the proposed increases have much larger effects: a reduction in 74.8 Mbbl/month for the current fee and 251.9 Mbbl/month for the \$1.00 per barrel fee. I attribute the difference to omitted variable bias: firm boundaries are positively correlated with distance, so failing to account for firm boundaries generates an upward bias in the shipment elasticity.

A significant degree of flexibility is required to correct for this bias. The lighter blue lines in Figure 5 are obtained using models that account for transaction costs less flexibly than the baseline specification. Model 1 corresponds to row 1. in Table 6, while Model 2 assumes infinite transaction costs for inactive relationships (see Section 5.1) and zero transaction costs otherwise.⁴³ These might be viewed as reasonable but parsimonious specifications given the evidence of transaction costs from the EQT/Rice merger in Section 2.2. Yet in each case, policy impacts are significantly overstated relative to the baseline model.

The estimates obtained here can be used to conduct benefit-cost analysis, given additional information: in particular, one must estimate the external costs of injection disposal. In this setting, the primary external costs derive from increased earthquake risk. Setting aside potential changes in extraction (see Section 5.3), other external costs include the social costs of freshwater depletion and potential trucking externalities such as air pollution, noise, and congestion. Measuring these costs is challenging. In prior work, Koster and van Ommeren (2015) find sizable impacts on housing prices from seismic activity linked to natural gas extraction in the Netherlands, while Muehlenbachs et al. (2021) find increases in traffic fatalities and car insurance premiums due to increased trucking in Pennsylvania.

7 Addressing transaction costs

In this section I investigate the relationship between transaction costs and injection disposal, and the implications of this relationship for the economic regulation of injection disposal.

Focusing on the baseline model, Figure 6 summarizes wastewater disposition with and without transaction costs. The leftmost scenario considers a counterfactual in which sharing

⁴³Hence, resources are assumed to be allocated efficiently among observed relationships. Assumptions of this kind are common when building equilibrium models in industrial organization.

is prohibited. In this scenario, most wastewater which is currently shared would instead be transferred to injection disposal facilities, with the remainder shifted to internal reuse. As a result, injection disposal increases 76% relative to the status quo. Hence, sharing appears to play an important role in limiting injection disposal under the status quo.

The previous finding is suggestive that policy interventions aimed at reducing transaction costs could potentially be useful tools for further reducing injection disposal. To quantify the potential gains from eliminating transaction costs altogether, the rightmost scenario considers a counterfactual with no transaction costs. In this case, injection disposal declines by 62%. From this perspective, policy interventions aimed at easing wastewater sharing could potentially have much larger impacts on injection disposal than would Ohio legislators' most aggressive proposals to date (see Section 6). Even modest interventions might have meaningful effects, particularly as a complement to injection disposal taxes.

Although in practice Ohio regulators may have limited ability to influence oil and gas regulations in Pennsylvania or other states, it is interesting to consider what types of interventions could be effective in reducing transaction costs. To this end, the third scenario from the left in Figure 6 presents a counterfactual in which transaction costs are eliminated only within the sending firm-receiving firm-months in which some sharing already occurs (i.e., within active relationships). Hence, transaction costs are eliminated only on the intensive margin of sharing. In this case injection disposal declines by 20%. This implies that the greater part of injection disposal attributable to transaction costs is due to frictions on the extensive margin of sharing. An implication is that policies aimed at reducing the costs of relationship formation may be especially effective in limiting injection disposal.

Two potentially important sources of friction in relationship formation are search and information costs on the one hand, and haggling and decision costs on the other (Dahlman, 1979). The first refers to the costs of identifying potential counterparties and evaluating gains from trade. The elevated transaction costs for cross-regional shipments reported in Table 7 are consistent with the presence of search and information costs. One way a state regulator can reduce search and information costs is by requiring complete and timely public disclosure of wastewater dispositions. Compared to Pennsylvania, other major oil and gas producing states such as Texas, Louisiana, New Mexico, and Oklahoma do not generally require detailed public disclosures of wastewater disposition. Another possibility is to require detailed reporting of all sources of base water in public completion reports. Based on my own analysis of Pennsylvania completion reports for the analysis period, source water reporting is often non-specific (e.g., "Produced Water") and sometimes incomplete.

Haggling and decision costs refer to the costs of writing contracts and bargaining over terms of trade. Groundwater Protection Council (2019) interviewed oil and gas firms operat-

ing throughout the country. One barrier to sharing highlighted in these interviews was legal and regulatory ambiguity regarding wastewater ownership, specifically on the question of whether wastewater is legally owned by the operating firm or the landowner. If wastewater is formally owned by landowners, “selling” wastewater may require consent of the landowner or royalty payments. This question currently remains unresolved in Pennsylvania.

A second issue highlighted in Groundwater Protection Council (2019) is legal and regulatory ambiguity surrounding liability for wastewater spills after wastewater crosses the firm boundary. If liability remains with the original operator, firms may be disincentivized from sharing, especially with riskier counterparties.⁴⁴ To assess the plausibility of this mechanism, I calculate for each firm the rate of spill-related PA DEP enforcement actions per barrel of wastewater over the analysis period, focusing on enforcement actions that resulted in monetary penalties. I then divide firms into “more frequent violators” and “less frequent violators” based on the median rate among the ten largest firms. Finally, I re-estimate the model interacting sender and receiver types with a firm boundary dummy. The results are reported in Table 9. Consistent with potential liability concerns, transaction costs are larger when the receiver is a more frequent violator but differ little when the sender is a more frequent violator. This suggests that clarifying liability could incentivize greater sharing (and reduce injection disposal), though such a policy could have other less desirable effects as well.

Interventions of this kind would be most appropriate if direct mechanisms for addressing injection disposal externalities (such as Pigouvian taxes) are difficult to implement. In contrast to an unpriced externality, the existence of transaction costs does not necessarily constitute a market failure. The central question is whether transaction costs are welfare-relevant: that is, whether they constitute “true” costs or simply “decision” costs (see, e.g., Bernheim and Rangel, 2009). If transaction costs simply capture inherent costs of production (such as the wages of staff who might spend time coordinating wastewater shipments), there is little rationale for policy intervention absent other market failures.

8 Conclusion

The Shale Boom lead to significant changes in the generation of wastewater from US oil and gas wells, raising questions about the sustainability of conventional disposal practices such as injection well disposal. Reuse is a promising alternative to conventional disposal. Focusing on the case of Pennsylvania, I develop an empirical framework that can be used to evaluate wastewater policy. The results show that the adoption of reuse may be constrained

⁴⁴Texas and Oklahoma have passed legislation to clarify liability in connection with reuse and disposal.

by frictions that make it difficult to transfer wastewater between firms. I use the model to evaluate the effects of disposal fees, showing that policy impacts appear modest once transaction costs are taken into account. Counterfactual simulations suggest that efforts to reduce transaction costs—in particular, efforts to reduce the costs of forming new relationships—are potentially a powerful complement to corrective taxation of injection disposal.

The analysis can be extended in a few ways. First, it would be useful to provide sharper evidence on the mechanisms that underly the estimated transaction costs. Without this evidence, it is difficult to recommend specific policy interventions to address transaction costs, though I find some evidence that supports general measures enhancing transparency and regulatory certainty. Second, it would be useful to investigate the robustness of the conclusions to the model assumptions, especially those pertaining to the cost shocks.

Pennsylvania has seen more extensive adoption of reuse than any other region. This is a consequence of two factors: relatively small volumes of wastewater, and a scarcity of injection disposal capacity. Recent years have seen a tightening of wastewater regulation in other regions such as the Permian Basin in Texas and New Mexico, where wastewater volumes are large and injection disposal capacity has historically been plentiful. The results of this paper highlight the importance of and potential benefits from accounting for transaction costs when designing wastewater policy in regions with less experience in reuse.

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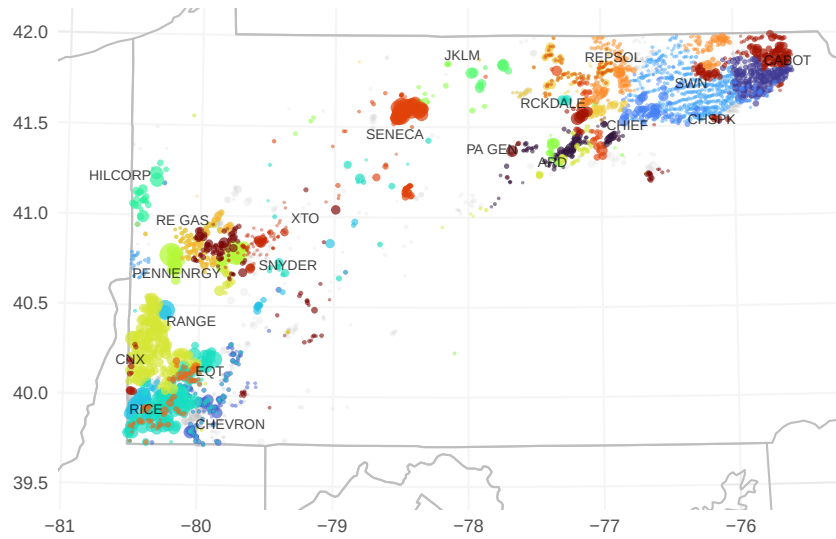
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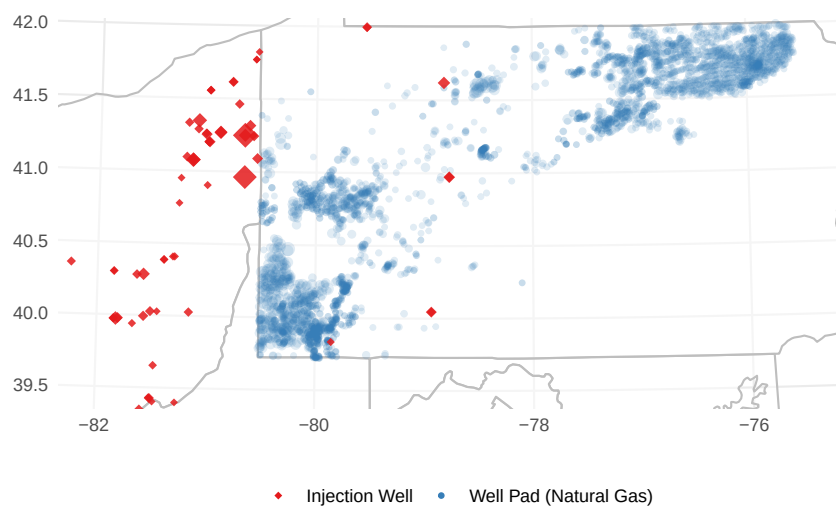
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Figure 1: Well Pad Locations for Twenty Largest Firms



Notes: Locations of well pads reporting wastewater disposition from Pennsylvania unconventional wells, 2017-2020. (1) Limited to twenty largest operators by wastewater volume. Each color corresponds to a different operator. (2) Point size corresponds to total disposal volume.

Figure 2: Injection Well and Well Pad Locations



Notes: Locations of injection wells receiving wastewater from Pennsylvania unconventional wells, 2017-2020, in comparison to gas wells reporting wastewater disposition. (1) For injection wells, marker size corresponds to total shipment volume. (2) Gray lines indicate state borders.

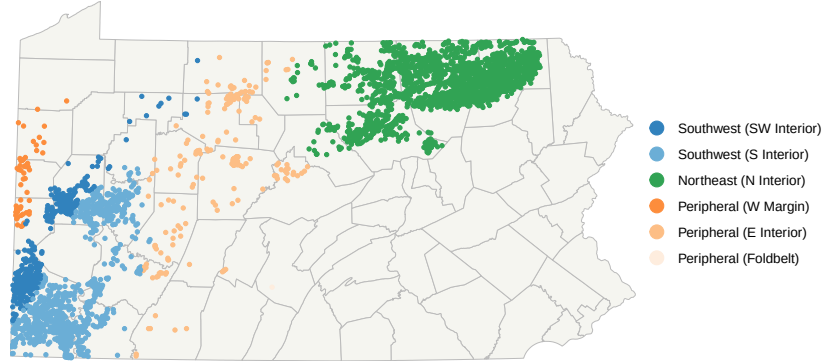
Table 1: Pennsylvania Unconventional Well Wastewater Disposition, 2017-2020

	Mbbl/Month	Percent
Total Disposal	4647.4	100.0
<u>Destination Type</u>		
Well pad or CTF	4120.3	88.7
Injection well	375.2	8.1
Other	151.8	3.3
<u>Well Pad or CTF Type</u>		
Internal	3174.5	68.3
Rival	387.6	8.3
3rd party	558.3	12.0

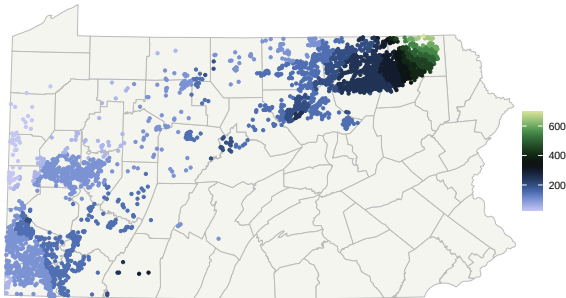
Notes: Author’s aggregation of Pennsylvania DEP waste report data. (1) Mbbl indicates one thousand oil barrels. (2) Classification of own and rival facilities based on identity of destination well pad operator at time of shipment. (3) 3rd party CTFs facilities are centralized wastewater treatment facilities not operated by oil and gas operators. (4) “Other” includes all shipments outside Pennsylvania (including shipments for reuse in West Virginia).

Figure 3: Key Geological Features of the Marcellus Shale, Wastewater Dispositions

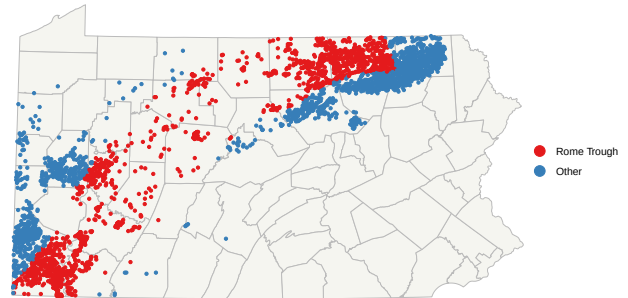
(a) Regions (USGS Assessment Units)



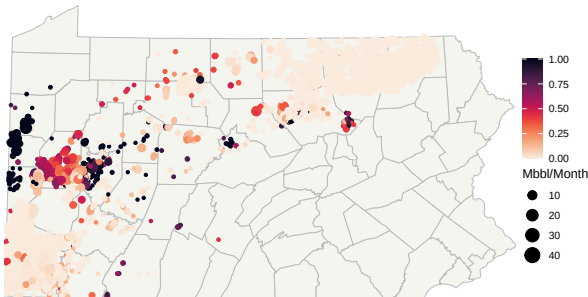
(b) Shale Thickness (in Feet)



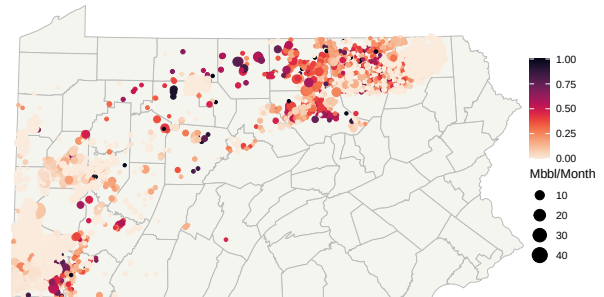
(c) Location of Rome Trough



(d) % Injection well



(e) % Rival well pad or CTF



Notes: Each dot represents a well pad appearing in the analysis sample. In (d) and (e), the size of each dot corresponds to total wastewater disposition. (a) USGS assessment unit from Higley (2019) indicated in parentheses. See also Higley et al. (2019). Author's aggregation. (b) Shale thickness from Energy Information Administration (2016a). (c) Rome Trough location from Energy Information Administration (2016b). (d) Percentage of wastewater disposition by well pad destined to injection well. (e) Percentage of wastewater disposition by well pad destined to rival well pad or CTF.

Table 2: Wastewater Disposition by Firm Type (Mbbl/Month)

	Southwest		Northeast		Peripheral
	Small	Large	Small	Large	
Total Disposal	493.4	2449.4	477.8	1108.2	118.7
<u>Destination Type</u>					
Well pad or CTF	306.5	2271.8	448.8	1077.8	15.5
Injection well	133.9	86.5	29.0	22.9	103.0
Other	53.0	91.0	0.1	7.5	0.2
<u>Well Pad or CTF Type</u>					
Internal	47.3	2006.5	202.9	917.6	0.2
Rival	100.5	65.8	153.8	54.8	12.7
3rd party	158.6	199.6	92.1	105.3	2.7
<u>Rival Type</u>					
Northeast (Large)	2.7	17.6	99.9	16.9	11.1
Northeast (Small)	0.4	0.4	52.7	33.7	0.1
Southwest (Large)	90.4	45.3	0.2	3.9	1.1
Southwest (Small)	6.8	2.2	0.2	0.1	0.4
Peripheral	-	0.1	-	0.0	-

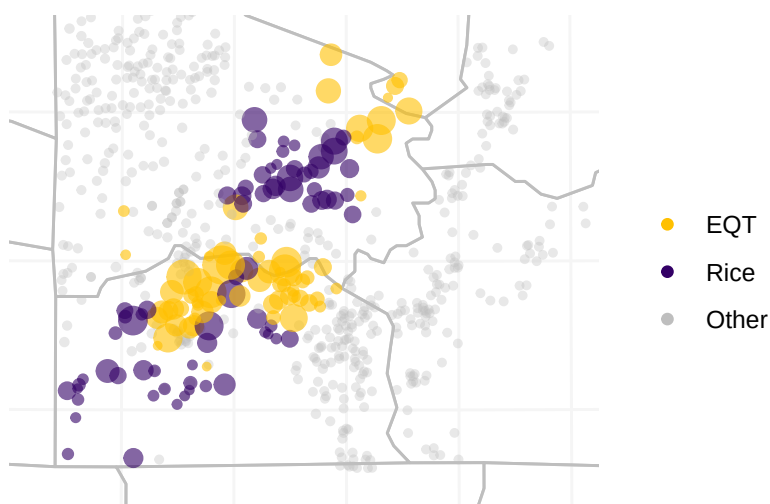
Notes: Author's aggregation of Pennsylvania DEP waste report data for unconventional wells, 2017-2020. For explanation of Destination and Well Pad/CTF Types, see notes to Table 1. (1) Mbbl indicates one thousand oil barrels. (2) Southwest (Northeast) includes S Interior and SW Interior (N Interior) assessment units from Higley et al. (2019). All other regions are Peripheral. (3) Large firm defined as one of the five largest firms within a region based on gas production during the analysis period.

Table 3: Distributions of Selected Statistics

	Mean	Std	5%	25%	50%	75%	95%
<i>Facility count per month</i>							
Well pads (origin)	1,712.6	76.6	1,587.2	1,659.8	1,707.5	1,763.0	1,831.0
Well pads (dest)	51.6	14.3	31.4	39.0	50.5	62.8	74.0
Producer CTFs (dest)	11.0	1.5	9.0	10.0	11.0	12.0	13.0
3rd party CTFs (dest)	10.9	1.3	9.0	10.0	11.0	12.0	13.0
<i>Truckloads sent or received by facility-month</i>							
Well pads (origin)	23.8	83.0	0.6	1.4	3.7	11.8	98.4
Well pads (dest)	430.8	915.4	0.9	4.0	29.5	363.0	2,347.7
Producer CTFs (dest)	905.1	1,429.1	2.9	51.4	271.8	1,071.6	4,683.4
3rd party CTFs (dest)	464.5	563.9	4.8	106.0	337.2	649.6	1,349.1
<i>Miles per truckload by destination type</i>							
Own pad or CTF	22.5	20.2	2.8	8.7	17.5	31.5	53.6
Rival pad or CTF	45.0	30.8	10.8	24.2	39.5	57.2	98.7
Injection well	75.5	54.0	18.1	30.1	68.0	88.6	215.9
3rd party CTF	31.4	29.8	4.4	10.6	24.4	44.6	76.0
All destinations	30.0	30.9	3.4	10.5	21.4	37.2	85.1

Notes: (1) Facility counts: Observations are facility type-months. For well pads (origin) an observation is the number of well pads recording wastewater disposal in a month. For well pads (dest) an observation is the number of well pads appearing as wastewater shipment destinations in a month. Producer CTFs (dest) and 3rd Party CTFs (dest) are defined similarly. (2) Truckloads sent or received: Observations are facility-months. For well pads (origin) an observation is the amount of wastewater shipped from a single well pad in month. For well pads (dest) an observation is the amount of wastewater shipped to a single well pad in a month. Producer CTFs (dest) and 3rd Party CTFs (dest) are defined similarly. A “truckload” is defined as 110 barrels of wastewater (i.e., 4,620 gallons). (3) Miles per truckload: Observations are origin-destination pairs. For own pad or CTF an observation is the distance between a well pad and an own pad or CTF to which it shipped wastewater. Remaining variables are defined similarly for different types of wastewater destinations.

Figure 4: EQT/Rice Pre-Merger Well Pad Locations



Notes: Locations of EQT and Rice well pads reporting wastewater disposition between 2017-01 and 2017-06. (1) For EQT/Rice well pads, point size corresponds to total disposal volume. (2) Gray dots indicate non-EQT/Rice wells reporting disposal. (3) Depicts Washington, Greene, and Fayette counties only. The prominent vertical and horizontal lines are state borders.

Table 4: Wastewater Disposition for Former EQT/Rice Well Pads (Mbbl/Month)

	EQT Well Pads		Rice Well Pads	
	Pre	Post	Pre	Post
Total Disposal	521.4	516.6	245.1	292.7
<u>Destination Type</u>				
Well pad or CTF	520.9	501.2	245.0	283.3
Injection well	-	6.4	0.1	2.9
Other	0.5	8.9	0.0	6.6
<u>Well Pad or CTF Type</u>				
Pre-Merger EQT	434.4	187.6	-	95.1
Pre-Merger Rice	-	40.4	173.1	38.2
Post-Merger EQT/Rice	-	213.3	-	134.6
Other Rival	46.5	15.4	-	1.4
3rd party	40.0	44.5	71.9	14.0

Notes: Author’s aggregation of Pennsylvania DEP waste report data for unconventional wells, 2017-2020. For explanation of Destination Types, see notes to Table 1. (1) Mbbl indicates one thousand oil barrels. (2) “Pre” refers to the period preceding the merger announcement, defined here as 2017-01 to 2017-06. “Post” refers to the period following merger consummation, defined here as 2017-11 to 2020-12. (3) EQT Well Pads (alternatively Pre-Merger EQT) refers to well pads from which EQT reported wastewater disposition during the “Pre” period. Rice Well Pads (alternatively Pre-Merger Rice) refers to well pads from which Rice reported wastewater disposition during the “Pre” period. Post-Merger EQT/Rice refers to well pads from which EQT first reported wastewater disposition subsequent to the merger. Note that Rice ceased reporting wastewater disposition after the merger.

Table 5: Preliminary Model Estimates

	Dependent Variable: # of Truckloads					
	(1)	(2)	(3)	(4)	(5)	(6)
Distance	-1.705*	-1.062*	-0.056*	-0.036*	-0.034*	-0.036*
	(0.017)	(0.017)	(0.001)	(0.001)	(0.001)	(0.001)
Firm Boundary		-4.921*		-4.958*		-4.910*
		(0.072)		(0.064)		(0.059)
Firm Boundary x Active Relationship					-2.342*	
					(0.177)	
EQT-Rice Post-Merger		0.017		-0.184*	-0.205*	-0.186*
		(0.039)		(0.033)	(0.063)	(0.033)
Distance Measure	Log Miles	Log Miles	Miles	Miles	Miles	Miles
κ -month FEs	Yes	Yes	Yes	Yes	Yes	Yes
δ -month FEs	Yes	Yes	Yes	Yes	Yes	Yes
Geologic Controls	No	No	No	No	No	Yes
Deviance	5.86×10^6	3.22×10^6	5.76×10^6	2.97×10^6	2.45×10^6	2.92×10^6
R^2	0.500	0.753	0.486	0.763	0.787	0.765
Observations	7,607,893	7,607,893	7,607,893	7,607,893	3,292,034	7,607,893

Notes: Parameter estimates for the cost of reuse. Significance levels: $*p < 0.05$. (1) A “truckload” of wastewater is defined as a quantity of 110 barrels. (2) Distance refers to the distance between origin well pad and destination facility. (3) Firm Boundary is an indicator for whether the destination is a Well Pad or CTF not operated by the reporting operator at the origin Well Pad. Active Relationship is an indicator for whether any sharing occurs within a sender firm-receiver firm-month. (4) EQT-Rice Post-Merger is a dummy for transactions in the period 2017-11 and 2020-12 for which the origin is a Pre-Merger EQT well pad the destination a Pre-Merger Rice well pad or vice versa. See notes to Table 4 for further details. (5) κ -month and δ -month FEs refers to origin and destination fixed effects. (6) Geological controls include interactions between origin and destination facility characteristics. Characteristics include: (i) USGS assessment unit (Higley et al., 2019); (ii) indicator for overlay of Rome Trough; (iii) indicator for overlay of region with Marcellus shale thickness in excess of 350 feet. (7) R^2 calculated as squared correlation between predicted and observed dependent variable. (8) Standard errors obtained by 2-step bootstrap (Zylkin, 2024).

Table 6: Analysis of Variance for the Transaction Costs of Sharing

Specification	Deviance	R_1^2	R_2^2
1. Distance Only	3.51×10^6	0.000	
2. Distance + Rival Dummy	1.39×10^6	0.604	0.000
3. Distance + Time $\times$ Rival Dummy	1.38×10^6	0.607	0.007
4. Distance + Rival $\times$ Firm Dummies	1.26×10^6	0.641	0.093
5. Distance + Rival $\times$ Firm Dummies (Directed)	1.22×10^6	0.653	0.123
6. Distance + Rival $\times$ Firm-Pair Dummies	1.14×10^6	0.676	0.180
7. Distance + Rival $\times$ Firm-Pair Dummies (Directed)	1.09×10^6	0.689	0.214
9. Distance + Time $\times$ Rival $\times$ Firm-Pair Dummies	9.41×10^5	0.732	0.323
10. Distance + Time $\times$ Rival $\times$ Firm-Pair Dummies (Directed)	9.21×10^5	0.738	0.337
11. Distance + Rival $\times$ Firm Sizes (Interacted)	1.38×10^6	0.607	0.006
12. Distance + Rival $\times$ Firm Geologic Regions (Interacted)	1.37×10^6	0.610	0.014
13. Distance + Rival $\times$ Firm Sizes and Geologic Regions (Interacted)	1.36×10^6	0.612	0.019

Notes: Deviance (shipments for reuse only) under alternative transaction cost specifications. (1) All models include distance in miles. (2) R_1^2 refers to the McFadden pseudo R^2 with specification 1. as the baseline (i.e., one minus the ratio of the reported deviance and the deviance for specification 1.). R_2^2 defined analogously with specification 2. as the baseline. (3) Time refers to Month, the frequency of the wastewater disposition data. (4) Firm Dummy specifications include indicators for whether each operating firm is party to a transaction. Firm Dummy (Directed) specifications include separate indicators depending on whether a firm is the origin or destination in a transaction. (5) Firm-Pair Dummy specifications include indicators for each relationship. Firm-Pair Dummy (Directed) specifications include separate indicators depending on the direction of trade within the relationship. (6) Firm Size defined by quartiles of the firm size distribution, as measured by gas production during the analysis period. (7) Geologic Region refers to USGS assessment unit (Higley et al., 2019).

Table 7: Estimated Firm Boundary Effects by Firm Region and Size

	To Southwest		To Northeast	
	Small	Large	Small	Large
From Northeast Large	182.8 (2.0)	157.6 (1.6)	107.3 (0.2)	163.0 (0.2)
From Northeast Small	192.6 (5.3)	221.6 (2.7)	122.5 (0.2)	136.5 (0.1)
From Southwest Large	157.2 (1.1)	150.1 (0.2)	208.6 (0.9)	127.9 (0.3)
From Southwest Small	134.4 (0.6)	131.2 (0.3)	207.3 (2.7)	192.6 (0.8)

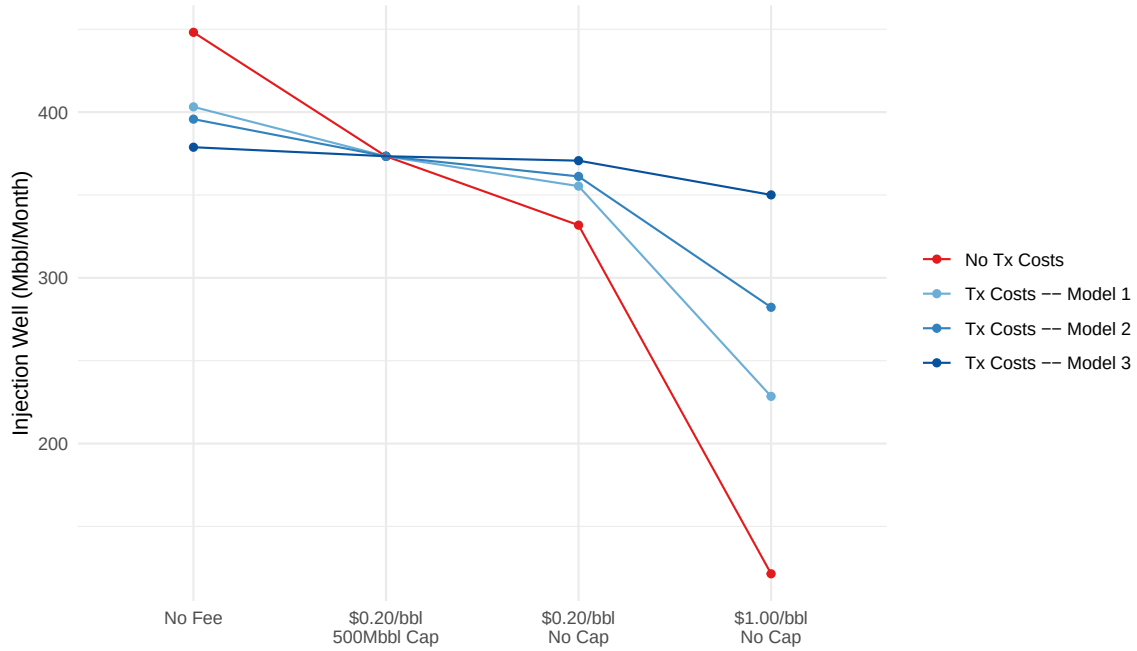
Notes: Parameter estimates obtained by interacting firm boundary dummy with Firm Region and Size classifications. (2) Southwest (Northeast) includes S Interior and SW Interior (N Interior) assessment units from Higley et al. (2019). All other regions are Peripheral. (3) Large firm defined as one of the five largest firms within a region based on gas production during the analysis period. (4) Peripheral region omitted for concision due to limited sample size. (5) Standard errors obtained by 2-step bootstrap (Zylkin, 2024).

Table 8: Quantification of Producer Surplus Gains from Reuse and Sharing

	Gas Production BCF/month	Wastewater Mbbl/month	Mean Shipment (mi)			Δ Surplus	
			No Reuse	Model	Δ	\$/bbl	\$/MMBtu
Northeast (Large)	181.3	1097.0	180.0	28.4	151.6	3.45	0.020
Northeast (Small)	53.1	475.2	180.2	39.0	141.2	3.21	0.028
Southwest (Large)	177.8	2333.3	94.3	25.6	68.7	1.56	0.020
Southwest (Small)	30.3	437.8	92.4	44.5	47.9	1.09	0.015
Peripheral	3.3	118.4	38.0	32.8	5.2	0.12	0.004
Total	445.9	4461.6	122.8	29.8	93.1	2.12	0.020

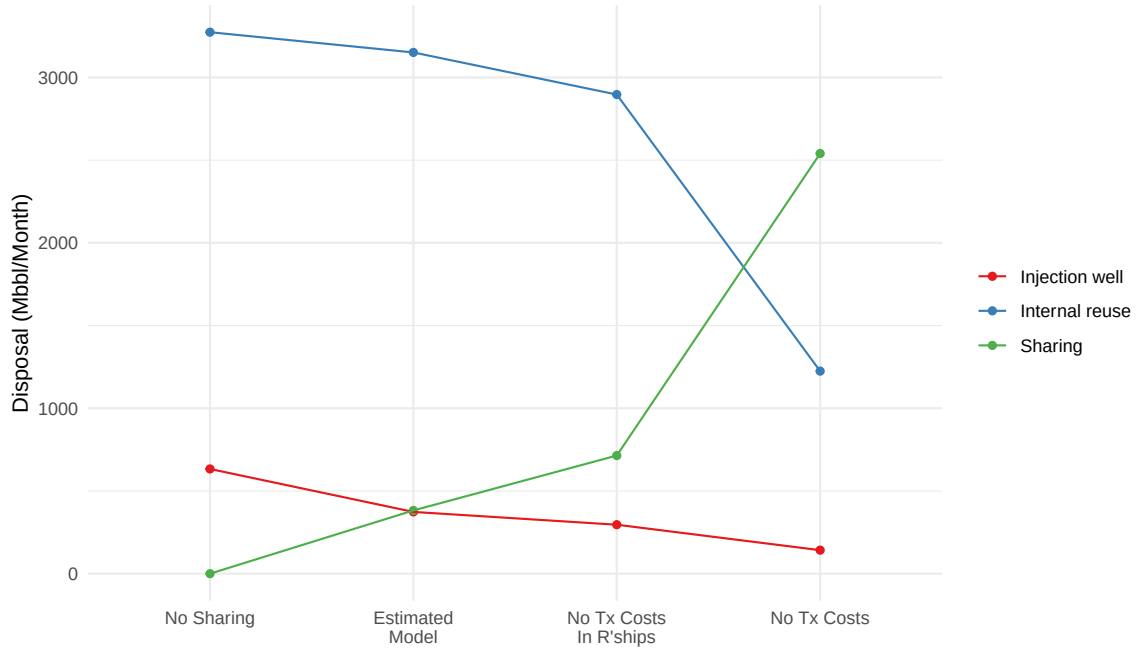
Notes: Counterfactual shipment distance and surplus estimates based on baseline model specification. (1) Gas production data from Pennsylvania DEP. (2) Southwest (Northeast) includes S Interior and SW Interior (N Interior) assessment units from Higley et al. (2019). All other regions are Peripheral. (3) Large firm defined as one of the five largest firms within a region based on gas production during the analysis period. (4) Surplus calculations assume the marginal cost of trucking is \$2.50 per mile. (5) BCF to MMBtu conversion assumes 1 Mcf $\approx$ 1.038 MMBtu.

Figure 5: Injection Disposal Rates Under Counterfactual Ohio Injection Disposal Taxes



Notes: Injection well disposal volumes for selected model specifications in fitted model and Ohio disposal tax counterfactuals. (1) Model specifications: No Tx Costs refers to distance-only model (specification 1. in Table 6). Tx Costs – Model 3 refers to baseline model with time-varying asymmetric relationship-specific transaction costs (specification 10. in Table 6). Tx Costs – Model 1 refers to the constant transaction costs model (specification 2. in Table 6). Tx Costs – Model 2 described in the text. (2) \$0.20/bbl 500 Mbbl Cap refers to existing Ohio injection disposal well tax. No Fee is a counterfactual with no tax. \$0.20/bbl No Cap and \$1.00/bbl No Cap are counterfactuals without the existing fee schedule kink at 500 Mbbl.

Figure 6: Wastewater Disposition Without Transaction Costs



Notes: Simulated wastewater disposition volumes for baseline model specifications in fitted model and selected counterfactuals. (1) Internal reuse refers to reuse at an operators own well pad or CTF. (2) 3rd party CTF disposition held fixed in all counterfactuals and omitted from figure. (3) No Sharing refers to a counterfactual with infinite transaction costs. (4) No Tx Costs in R'ships refers to a counterfactual with zero transaction costs among sending firm-receiving firm-months with observed sharing and infinite transaction costs for all other potential sharing transactions. (5) No Tx Costs refers to a counterfactual with no transaction costs for any potential sharing transactions.

Table 9: Firm Boundary Effects by Frequency of Spill-Related Enforcement Actions

	To Less Freq. Violator	To More Freq. Violator
From Less Frequent Violator	123.3 (0.1)	162.5 (0.2)
From More Frequent Violator	124.4 (0.1)	176.2 (0.3)

Notes: Parameter estimates obtained by interacting firm boundary dummy with the frequency of spill-related DEP enforcement actions that resulted in monetary penalties during the analysis period. (1) For each firm, I calculate the rate of enforcement actions per barrel of wastewater. More Frequent Violators include a mix of large and small firms with violation rates exceeding the median rate among the ten largest firms. All other firms are Less Frequent Violators. (2) Standard errors obtained by 2-step bootstrap (Zylkin, 2024).

A Appendices

A.1 Data preparation

Classification of reuse and sharing The main dataset consists of Oil and Gas Well Waste Reports collected from the Pennsylvania Department of Environmental Protection web site.⁴⁵ Operators are required to report the method of disposal for various waste products, including solids such as drill cuttings and shredded containment liners. I rely on the classifications from Wunz (2015) as well as internet research on the functions performed at different waste facilities (e.g., landfills vs. injection wells) to identify presumably reusable wastewater. This procedure is inevitably imperfect. Reporting errors are possible, and not all liquid waste necessarily represents reusable wastewater.⁴⁶

As described in the main text, the waste reports do not report the dates or quantities associated with specific transfer events, but rather the aggregate quantities of different types of waste transferred from a given well to a given disposal location during a month. Wastewater intended for reuse can be transferred either to a CTF prior to reuse or directly to another well pad for reuse. These cases appear differently in the data. In the former case, it is not possible to identify the ultimate location of reuse. However, whether the treatment facility is operated by the reporting firm or by a third party can be inferred from the reported permit information and facility names (although in some cases this requires consulting separate DEP resources). If the destination is a well pad located in Pennsylvania, as occurs most often, a numeric identifier associated with the destination well pad is provided. I use this numeric identifier to determine whether a given amount of wastewater was transferred for internal or external reuse. In particular, I classify reuse locations as internal or external depending on whether the reporting firm is currently listed as an operator for any well at the destination well pad (in a separate DEP data source). If the destination well pad is located outside of Pennsylvania (primarily in West Virginia), no such identifier is provided, and I do not attempt to infer the ownership of the destination well pad. I identify firms by their DEP OGO Number (where OGO is an acronym for “Oil and Gas Operator”). I rely on press releases and changes in the reporting operator over time to track changes in corporate ownership (the Rice-EQT merger was the most significant but not the sole merger during the sample period). It is rare for multiple operators to be associated with the same well pad, but when this is the case I treat the well pad as “internal” for both parties.

Typically several wells are located at a single well pad, which encompasses common

⁴⁵PA DEP (2021). Other referenced data sources include GWPC and IOGCC (2021), SRBC (2021), Ohio DNR (2024).

⁴⁶For instance, sludges produced as a byproduct of treatment for reuse are often sent to injection wells.

infrastructure such as access roads and storage tanks. Technically operators are required to report waste information on a well-by-well basis, but because wastewater is often stored in a single location on the pad many simply report well pad-level averages. Therefore I focus on the well pad rather than the well as my primary unit of analysis. I infer the number of shipments in a month by dividing the total volume by the capacity of a typical water hauling truck.⁴⁷ To mitigate the impact of data reporting errors, I winsorize shipment volumes at the 99.9%-tile (about 77,000 barrels, or 600-700 truckloads in a month).

Analysis period For the main analysis, I consider waste reports for all unconventional wells and for all production periods between January 2017 and December 2020. This choice of analysis period reflects the fact that the waste reporting format was modified in January 2017 to consistently indicate the location of reuse.⁴⁸ Prior to January 2017, it is possible to determine whether reuse occurred, but not whether reuse occurred internally or via sharing.

Distance calculation I calculate over-the-road shipment distances using the Open Source Routing Machine (Luxen and Vetter, 2011) and data from OpenStreetMaps.⁴⁹

A.2 Validation of Freshwater Usage Rates

$\hat{\mu}_{0\delta}$ is constructed as the sum of observed wastewater shipments divided by an estimated firm-year level wastewater consumption rate. The latter is defined as the difference between the sum of “TotalBaseFluid” for all fracking events appearing in the FracFocus data and observed disposals. A well’s freshwater usage is equal to one minus its wastewater usage. Freshwater usage is observable for a subset of wells in Northeastern Pennsylvania from the SRBC data. For firm-years appearing in the SRBC data, Figure 7 plots (i) the calculated wastewater usage rate based on the FracFocus “TotalBaseFluid” and disposal records against on the x-axis and (ii) the true wastewater usage rate in the SRBC data on the y-axis. An observation is a firm-year. A linear regression yields a coefficient 1.13 (SE 0.2) and an R^2 value of 0.49.

A.3 Trucking cost assumptions

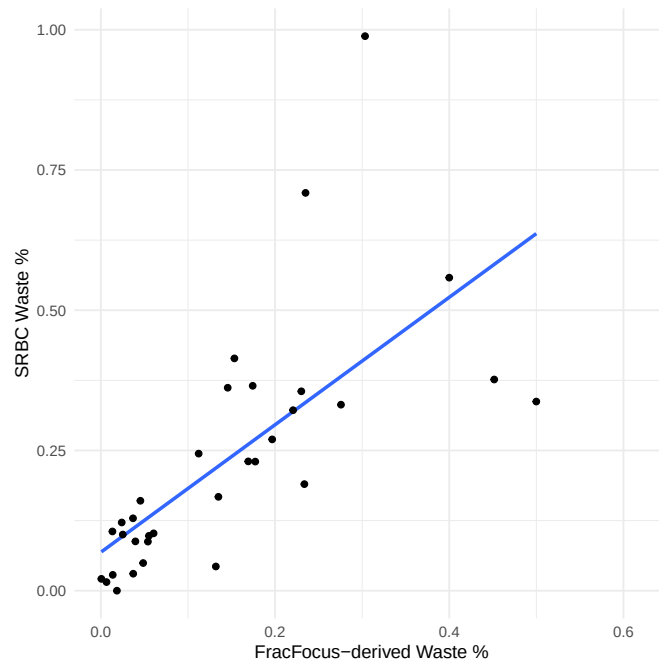
My preferred estimate is based on the following assumptions:

⁴⁷Line items in the data are frequently reported in integer multiples of a truck capacities ranging between 80 and 130 barrels.

⁴⁸I choose to retain data from the COVID pandemic period. Although drilling rates in general fell during this period, the demand for disposal did not, and the overall reuse rate remained stable.

⁴⁹I do not account for roadway-specific vehicle weight or hazardous material restrictions that could alter optimal shipment routes for wastewater-hauling trucks in comparison to passenger vehicles.

Figure 7: Predicted vs. Observed Wastewater Percentage in SRBC Sample



Notes: (1) Observations are firm-years. (2) Blue line is an OLS regression line. (3) SRBC Waste % is the average reuse rate of a firm's wells among wells appearing in the SRBC sample of well-level water usage in Northeastern Pennsylvania. (4) FracFocus-derived Waste % is described in Appendix A.2.

1. Trucker wages are assumed to be \$22.62 per hour based on the BLS estimate of the average hourly wage of Pennsylvania truckers in 2018.
2. Average speed is assumed to be 27.8 miles per hour based on the ratio of duration and distance along calculated optimal routes for observed shipments.
3. Tanker-truck fuel efficiency is assumed to be 13.5 gallons of gasoline equivalent per 100 miles unladen and 15.4 gallons per 100 miles laden (assuming vehicle weights of 30,000 and 70,000 pounds, respectively) based on Bureau of Transportation Safety figures. Assuming 50% utilization this implies fuel efficiency is 6.9 miles per gallon of gasoline, equivalent to 7.9 miles per gallon of diesel.⁵⁰
4. Diesel price is assumed to be \$3.361 per gallon based on the 2018 average diesel retail price reported by EIA for PADD1B (East Coast - Mid-Atlantic).

A.4 Estimation of Ohio disposal fees

I obtained well-month level injection disposal data from the Ohio DNR. However, the data does not distinguish between in-district and out-of-district brines. I therefore consider two possible approaches to calculating average fees at the well level. One possibility is assume that 100% of observed disposal originates out-of-district. Another is to assume that only shipments observed in the Pennsylvania disposal data qualify as out-of-district disposals (thus assuming that the Ohio injection wells used by Pennsylvania firms receive no shipments from West Virginia or elsewhere). These measures coincide whenever I observe more than 500 Mbbl in annual shipments from Pennsylvania well pads to a particular injection well. Both measures produce similar (quantity-weighted) average fees: \$0.116 per barrel using the first approach, and \$0.135 per barrel using the second. Because the difference is modest, I adopt the first approach because it produces more conservative estimates of the impacts from the current disposal fee.

⁵⁰Unlike dry van trucking, backhauls for produced water are rare due to the necessity of cleaning the tank between loads.